

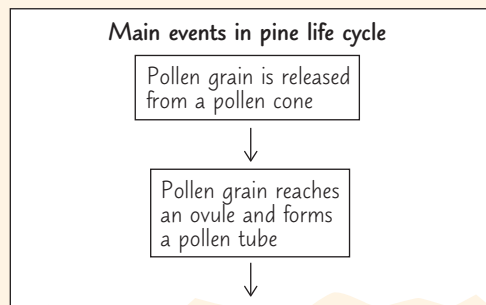
30 Plant Diversity II: The Evolution of Seed Plants

KEY CONCEPTS

- 30.1** Seeds and pollen grains are key adaptations for life on land p. 637
- 30.2** Gymnosperms bear “naked” seeds, typically on cones p. 640
- 30.3** The reproductive adaptations of angiosperms include flowers and fruits p. 644
- 30.4** Human welfare depends on seed plants p. 651

Study Tip

Make flowcharts: Make simple flowcharts showing the main events in the life cycles of a pine and flowering plant. The example shown is the beginning of a flowchart summarizing the pine life cycle.



➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 30
- Figure 30.3 Walkthrough: From Ovule to Seed in a Gymnosperm
- Figure 30.12 Walkthrough: The Life Cycle of an Angiosperm
- Animation: Angiosperm Life Cycle

For Instructors to Assign (in Item Library)

- Scientific Skills Exercise: Using Natural Logarithms to Interpret Data
- Tutorial: Gymnosperms



Figure 30.1 In 1980, Mount St. Helens in Washington state erupted, leaving the region covered in ash and devoid of visible life. Just a few years later, seed plants such as fireweed (*Chamerion angustifolium*) had already colonized the barren landscape—an example of seed plants’ versatility and success as the dominant producers on land.



Fireweed seed

What adaptations have enabled seed plants to make up the vast majority of plant biodiversity?

Reduced male and female gametophytes

develop within parental sporophytes, protected from environmental stresses.

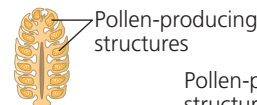
In gymnosperms, the gametophytes develop within cones.



In angiosperms, the gametophytes develop within flowers.



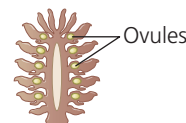
Pollen protects male gametophytes (which produce sperm) and can be transported by wind or animals.



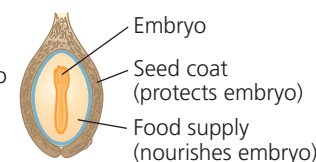
Pollen-producing structures



Ovules protect female gametophytes (which produce eggs).



An ovule fertilized by pollen develops into a **seed**.



Fireweed and other early arrivals reached the blast zone in Figure 30.1 as seeds. A **seed** consists of an embryo and its food supply, surrounded by a protective coat. When mature, seeds are dispersed from their parent by wind or other means, enabling them to colonize distant locations.

Plants not only have affected the recovery of regions such as Mount St. Helens but also have transformed Earth. Continuing the saga of how this occurred, this chapter follows the emergence and diversification of the group to which fireweed belongs, the seed plants.

CONCEPT 30.1

Seeds and pollen grains are key adaptations for life on land

We begin with an overview of terrestrial adaptations that seed plants added to those already present in nonvascular plants (bryophytes) and seedless vascular plants (see Concept 29.1). In addition to seeds, all seed plants have reduced gametophytes, heterospory, ovules, and pollen. As we'll see, these adaptations helped

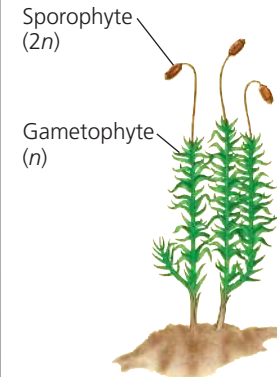
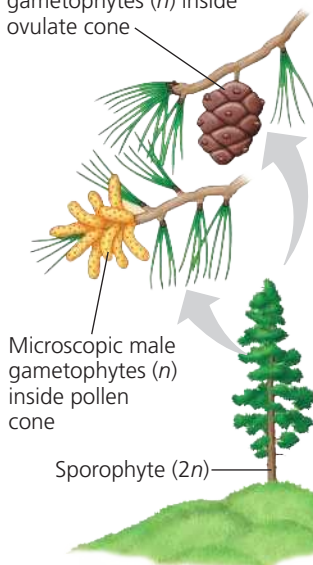
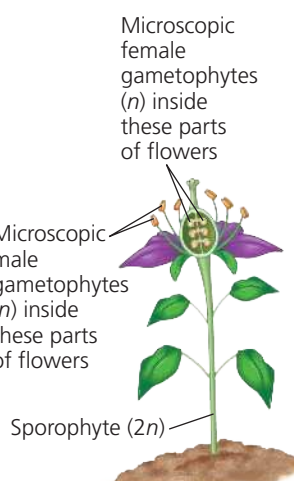
seed plants cope with conditions such as drought and exposure to ultraviolet (UV) radiation in sunlight. They also freed seed plants from requiring water for fertilization, enabling reproduction under a broader range of conditions than in seedless plants.

Advantages of Reduced Gametophytes

Mosses and other bryophytes have life cycles dominated by gametophytes, whereas ferns and other seedless vascular plants have sporophyte-dominated life cycles. The evolutionary trend of gametophyte reduction continued further in the vascular plant lineage that led to seed plants (Figure 30.2). While the gametophytes of seedless vascular plants are visible to the naked eye, the gametophytes of most seed plants are microscopic.

This miniaturization allowed for an important evolutionary innovation in seed plants: Their tiny gametophytes can develop from spores retained within the sporangia of the parental sporophyte. This arrangement can protect the gametophytes from environmental stresses. For example, the moist reproductive tissues of the sporophyte shield the gametophytes from UV radiation and protect them from drying out. This relationship

▼ **Figure 30.2** Gametophyte-sporophyte relationships in different plant groups.

	PLANT GROUP		
	Mosses and other nonvascular plants	Ferns and other seedless vascular plants	Seed plants (gymnosperms and angiosperms)
Gametophyte	Dominant	Reduced, independent (photosynthetic and free-living)	Reduced (usually microscopic), dependent on surrounding sporophyte tissue for nutrition
Sporophyte	Reduced, dependent on gametophyte for nutrition	Dominant	Dominant
Example			Gymnosperm  Angiosperm 

MAKE CONNECTIONS In seed plants, how might retaining the gametophyte within the sporophyte affect embryo fitness? (See Concepts 17.5, 23.1, and 23.4 to review mutagens, mutations, and fitness.)

also enables the developing gametophytes to obtain nutrients from the parental sporophyte. In contrast, the free-living gametophytes of seedless vascular plants must fend for themselves.

Heterospory: The Rule Among Seed Plants

You read in Concept 29.3 that most seedless plants are *homosporous*—they produce one kind of spore, which usually gives rise to a bisexual gametophyte. Ferns and other close relatives of seed plants are *homosporous*, suggesting that seed plants had homosporous ancestors. At some point, seed plants or their ancestors became *heterosporous*, producing two kinds of spores: Megasporangia on modified leaves called megasporophylls produce *megaspores* that give rise to female gametophytes, and microsporangia on modified leaves called microsporophylls produce *microspores* that give rise to male gametophytes. Each megasporangium has one megaspore, whereas each microsporangium has many microspores.

As noted previously, the miniaturization of seed plant gametophytes probably contributed to the great success of this clade. Next, we'll look at the development of the female gametophyte within an ovule and the development of the male gametophyte in a pollen grain. Then we'll follow the transformation of a fertilized ovule into a seed.

Ovules and Production of Eggs

Although a few species of seedless plants are heterosporous, seed plants are unique in retaining the megasporangium within the parent sporophyte. A layer of sporophyte tissue called **integument** envelops and protects the megasporangium. Gymnosperm megasporangia are surrounded by one integument,

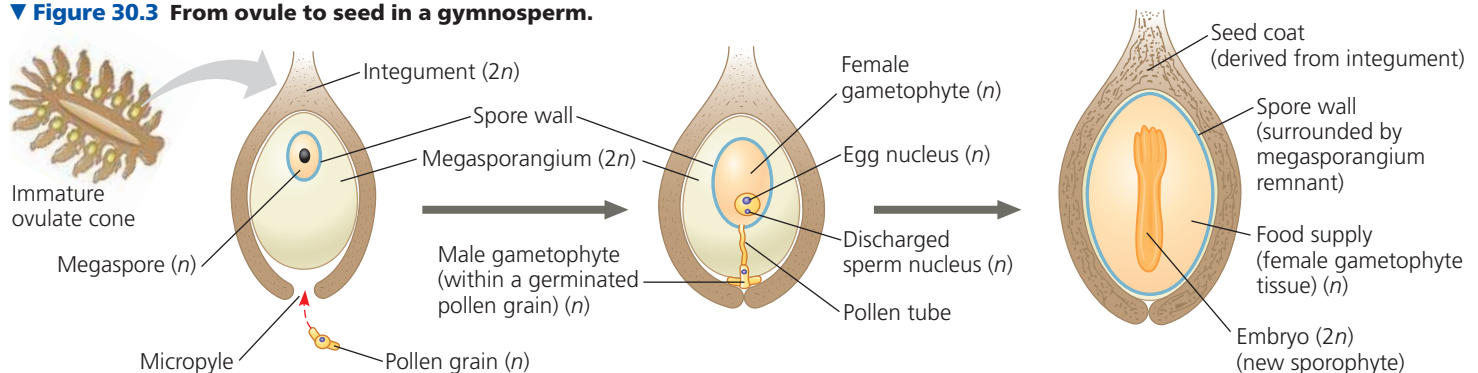
whereas those in angiosperms usually have two integuments. The whole structure—megasporangium, megaspore, and their integument(s)—is called an **ovule** (Figure 30.3a). Inside each ovule (from the Latin *ovulum*, little egg), a female gametophyte develops from a megaspore and produces one or more eggs.

Pollen and Production of Sperm

A microspore develops into a **pollen grain** that consists of a male gametophyte enclosed within the pollen wall. (The wall's outer layer is made of molecules secreted by sporophyte cells, so we refer to the male gametophyte as being *in* the pollen grain, not *equivalent to* the pollen grain.) Sporopollenin in the pollen wall protects the pollen grain as it is transported by wind or by hitchhiking on an animal. The transfer of pollen to the part of a seed plant that contains the ovules is called **pollination**. If a pollen grain germinates (begins growing), it gives rise to a pollen tube that discharges sperm into the female gametophyte within the ovule, as shown in Figure 30.3b.

In nonvascular plants and seedless vascular plants such as ferns, free-living gametophytes release flagellated sperm that swim through a film of water to reach eggs. Given this requirement, it is not surprising that many of these species live in moist habitats. But a pollen grain can be carried by wind or animals, eliminating the dependence on water for sperm transport. The ability of seed plants to transfer sperm without water likely contributed to their colonization of dry habitats. The sperm of seed plants also do not require motility because they are carried to the eggs by pollen tubes. The sperm of some gymnosperm species (such as cycads and ginkgos, shown in Figure 30.7) retain the ancient flagellated condition, but flagella have been lost in the sperm of most gymnosperms and all angiosperms.

▼ **Figure 30.3** From ovule to seed in a gymnosperm.



(a) Unfertilized ovule. In this longitudinal section through the ovule of a pine (a gymnosperm), a fleshy megasporangium is surrounded by a protective layer of tissue called an integument. The micropyle, the only opening through the integument, allows entry of a pollen grain.

(b) Fertilized ovule. A megaspore develops into a female gametophyte, which produces an egg. The pollen grain, which had entered through the micropyle, contains a male gametophyte. The male gametophyte develops a pollen tube that discharges sperm, thereby fertilizing the egg.

(c) Gymnosperm seed. Fertilization initiates the transformation of the ovule into a seed, which consists of a sporophyte embryo, a food supply, and a protective seed coat derived from the integument. The megasporangium dries out and collapses.

VISUAL SKILLS Based on this figure, a gymnosperm seed contains cells from how many different plant generations? Identify the cells and whether each is haploid or diploid.

➔ **Mastering Biology Figure Walkthrough**

Scientific Skills Exercise

Using Natural Logarithms to Interpret Data

How Long Can Seeds Remain Viable in Dormancy?

Environmental conditions can vary greatly over time, and they may not be favorable for germination when seeds are produced. One way that plants cope with such variation is through seed dormancy. Under favorable conditions, seeds of some species can germinate after many years of dormancy.

One unusual opportunity to test how long seeds can remain viable occurred when seeds from date palm trees (*Phoenix dactylifera*) were discovered under the rubble of a 2,000-year-old fortress near the Dead Sea. As you saw in the Chapter 2 Scientific Skills Exercise and Concept 25.2, scientists use radiometric dating to estimate the ages of fossils and other old objects. In this exercise, you will estimate the ages of three of these ancient seeds by using natural logarithms.

How the Experiment Was Done Scientists measured the fraction of carbon-14 that remained in three ancient date palm seeds: two that were not planted and one that was planted and germinated. For the germinated seed, the scientists used a seed coat fragment found clinging to a root of the seedling. (The seedling grew into the plant in the photo.)

Data from the Experiment This table shows the fraction of carbon-14 remaining from the three ancient date palm seeds.

	Fraction of Carbon-14 Remaining
Seed 1 (not planted)	0.7656
Seed 2 (not planted)	0.7752
Seed 3 (germinated)	0.7977



INTERPRET THE DATA

A logarithm is the power to which a base is raised to produce a given number x . For example, if the base is 10 and $x = 100$, the logarithm of 100 equals 2 (because $10^2 = 100$). A natural logarithm ($\ln$) is the logarithm of a number x to the base e , where e is about 2.718. Natural logarithms are useful in calculating rates of some natural processes, such as radioactive decay.

The equation $F = e^{-kt}$ describes the fraction F of an original isotope remaining after a period of t years; the exponent is negative because it refers to a decrease over time. The constant k provides a measure of how rapidly the original isotope decays. For the decay of carbon-14 to nitrogen-14, $k = 0.00012097$.

To estimate t , the age of the three seeds, we rearrange the equation $F = e^{-kt}$ to find an equation for t :

$$t = -\left(\frac{\ln F}{k}\right)$$

- Using the equation for t , the data from the table, and a calculator, estimate the ages of seed 1, seed 2, and seed 3.
- Why do you think there was more carbon-14 in the germinated seed?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Data from S. Sallon et al., Germination, genetics, and growth of an ancient date seed, *Science* 320:1464 (2008).

The Evolutionary Advantage of Seeds

If a sperm fertilizes an egg of a seed plant, the zygote grows into a sporophyte embryo. As shown in **Figure 30.3c**, the ovule develops into a seed: the embryo, with a food supply, packaged in a protective coat derived from the integument(s).

Seeds provide protection from harsh conditions and facilitate dispersal to new habitats—as do the spores of seedless plants. Moss spores, for example, may survive even if the local environment becomes too cold, too hot, or too dry for the mosses themselves to live. Their tiny size enables the spores to be dispersed in a dormant state to a new area, where they can germinate and give rise to new moss gametophytes if and when conditions are favorable enough for them to break dormancy. Spores were the main way that mosses, ferns, and other seedless plants spread over Earth for the first 100 million years of plant life on land.

Although mosses and other seedless plants continue to be very successful today, seeds represent a major evolutionary innovation that contributed to the opening of new ways of life for seed plants. What advantages do seeds provide over spores? Spores are usually single-celled, whereas seeds

are multicellular, consisting of an embryo protected by a layer of tissue, the seed coat. A seed can remain dormant for days, months, or even years after being released from the parent plant, whereas most spores have shorter lifetimes. Also, unlike spores, seeds have a supply of stored food. Most seeds land close to their parent sporophyte plant, but some are carried long distances (up to hundreds of kilometers) by wind or animals. If conditions are favorable where it lands, the seed can emerge from dormancy and germinate, with its stored food providing critical support for growth as the sporophyte embryo emerges as a seedling. As we explore in the **Scientific Skills Exercise**, some seeds have germinated after more than 1,000 years.

CONCEPT CHECK 30.1

- Contrast how sperm reach the eggs of seedless plants with how sperm reach the eggs of seed plants.
- What features not present in seedless plants have contributed to the success of seed plants on land?
- WHAT IF?** If a seed could not enter dormancy, how might that affect the embryo's transport or survival?

For suggested answers, see Appendix A.

CONCEPT 30.2

Gymnosperms bear “naked” seeds, typically on cones



Nonvascular plants (bryophytes)
Seedless vascular plants
Gymnosperms
Angiosperms

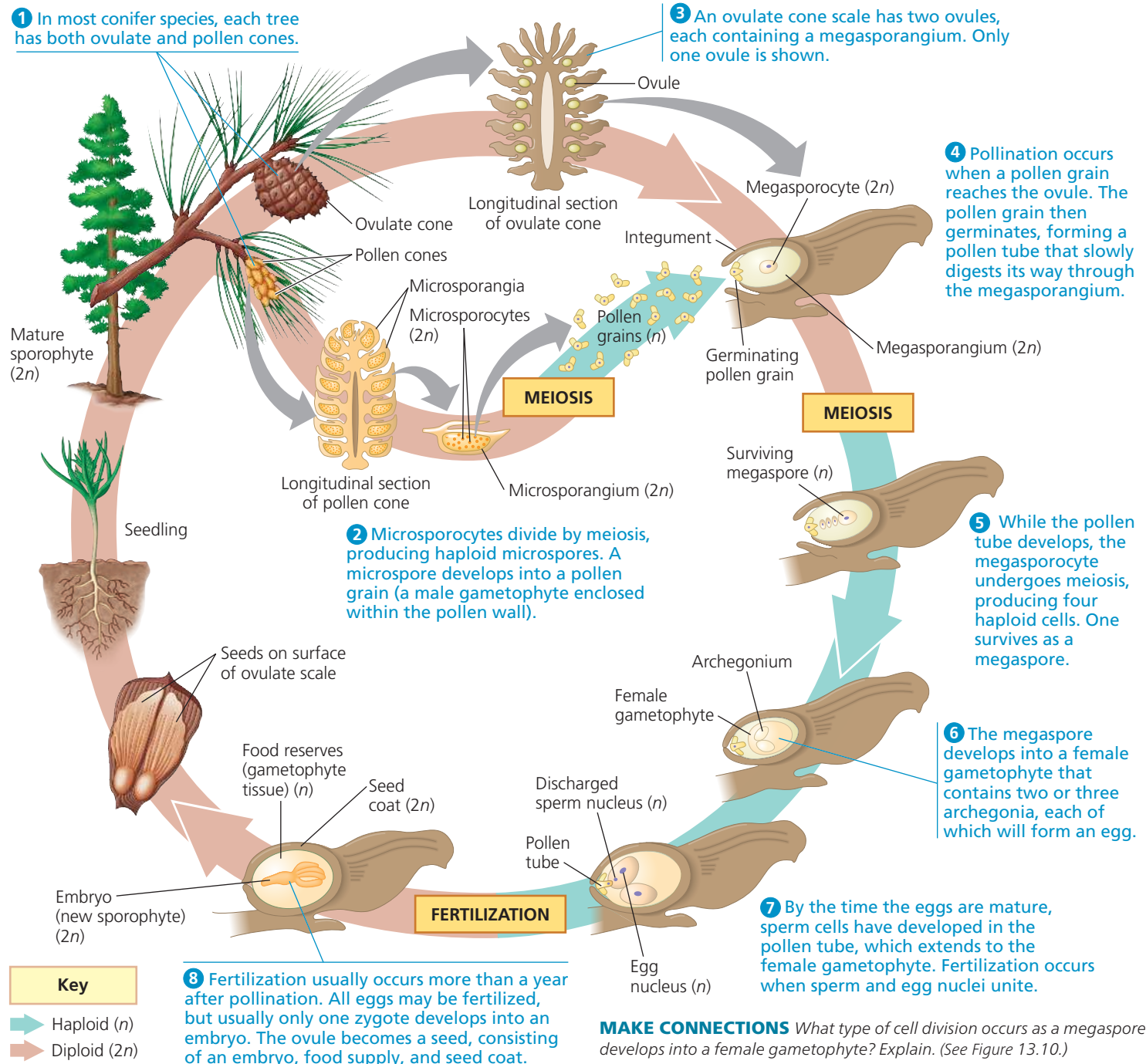
Extant seed plants form two sister clades: gymnosperms and angiosperms. Recall that gymnosperms have “naked” seeds exposed on sporophylls that usually form cones. (Angiosperm seeds are enclosed in chambers that mature

into fruits.) Most gymnosperms are cone-bearing plants called **conifers**, such as pines, firs, and redwoods.

The Life Cycle of a Pine

As you read earlier, seed plant evolution has included three key reproductive adaptations: the miniaturization of their gametophytes; the advent of the seed as a resistant, dispersible stage in the life cycle; and the appearance of pollen as an airborne agent that brings gametes together. **Figure 30.4** shows how these adaptations come into play during the life cycle of a pine, a familiar conifer.

▼ **Figure 30.4** The life cycle of a pine.



The pine tree is the sporophyte; its sporangia are located on scalelike structures packed densely in cones. Like all seed plants, conifers are heterosporous. As such, they have two types of sporangia that produce two types of spores: microsporangia that produce microspores and megasporangia that produce megaspores. In conifers, the two types of spores are produced by separate cones: small pollen cones and large ovulate cones.

Pollen cones have a relatively simple structure: Their scales are modified leaves (microsporophylls) that bear microsporangia. Within each microsporangium, cells called microsporocytes undergo meiosis, producing haploid microspores. Each microspore develops into a pollen grain containing a male gametophyte. In conifers, the yellow pollen is released in large amounts and carried by the wind, dusting everything in its path.

Ovulate cones are more complex: Their scales are compound structures composed of both modified leaves (megasporephylls bearing megasporangia) and modified stem tissue. Within each megasporangium, megasporocytes undergo meiosis and produce haploid megaspores inside the ovule. Surviving megaspores develop into female gametophytes, which are retained within the sporangia.

In most pine species, each tree has both types of cones. From the time pollen and ovulate cones appear on the tree, it takes nearly three years for the male and female gametophytes to be produced and brought together and for mature seeds to form from fertilized ovules. The scales of each ovulate cone then separate, and seeds are dispersed by the wind. A seed that lands in a suitable environment germinates, its embryo emerging as a pine seedling.

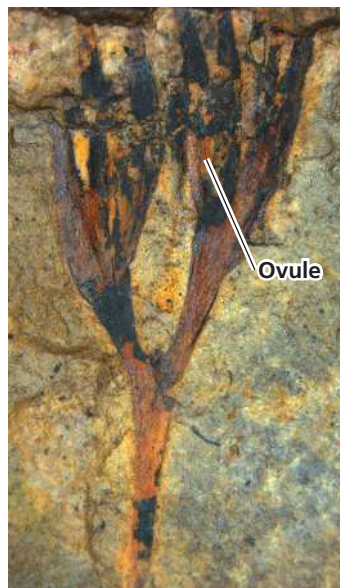
Early Seed Plants and the Rise of Gymnosperms

The origins of characteristics found in pines and other living seed plants date back to the late Devonian period (380 million years ago). Fossils from that time reveal that some plants had acquired features that are also present in seed plants, such as megaspores and microspores. For example, *Archaeopteris* was a heterosporous tree with a woody stem. It grew up to 20 m tall and had fernlike leaves. But it did not bear seeds and therefore is not classified as a seed plant.

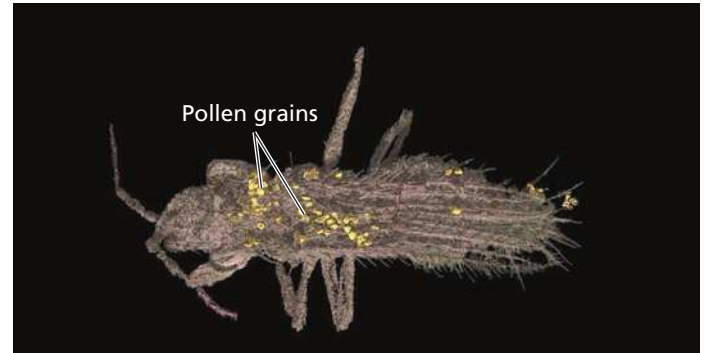
The earliest evidence of seed plants comes from 360-million-year-old fossils of plants in the genus *Elkinsia* (Figure 30.5). These and other early seed plants lived 55 million years before the first fossils classified as gymnosperms and more than 200 million years before the first fossils of angiosperms. These early seed plants became extinct, and we don't know which extinct lineage gave rise to the gymnosperms.

The oldest fossils of species from an extant lineage of gymnosperms are 305 million years old. These early gymnosperms lived in moist Carboniferous ecosystems that were

▼ **Figure 30.5** A fossil of the early seed plant *Elkinsia*.



▼ **Figure 30.6** An ancient pollinator. This 110-million-year-old fossil shows pollen on an insect, the thrip *Gymnopolisthrips minor*. Structural features of the pollen suggest that it was produced by gymnosperms (most likely by species related to extant ginkgos or cycads). Although most gymnosperms today are wind-pollinated, many cycads are insect-pollinated.



dominated by lycophytes, horsetails, ferns, and other seedless vascular plants. As the Carboniferous period gave way to the Permian (299 to 252 million years ago), the climate became much drier. As a result, the lycophytes, horsetails, and ferns that dominated Carboniferous swamps were largely replaced by gymnosperms, which were better suited to the drier climate.

Gymnosperms thrived as the climate dried, in part because they have the key terrestrial adaptations found in all seed plants, such as seeds and pollen. In addition, some gymnosperms were particularly well suited to arid conditions because of the thick cuticles and relatively small surface areas of their needle-shaped leaves.

Gymnosperms dominated terrestrial ecosystems throughout much of the Mesozoic era, which lasted from 252 to 66 million years ago. In addition to serving as the food supply for giant herbivorous dinosaurs, these gymnosperms were involved in many other interactions with animals. Recent fossil discoveries, for example, show that some gymnosperms were pollinated by insects more than 100 million years ago—the earliest evidence of insect pollination in any plant group (Figure 30.6).

Late in the Mesozoic, angiosperms began to replace gymnosperms in some ecosystems.

Gymnosperm Diversity

Although angiosperms now dominate most terrestrial ecosystems, gymnosperms remain an important part of Earth's flora. For example, vast regions in northern latitudes are covered by forests of conifers (see Figure 52.13).

Of the ten plant phyla (see Table 29.1), four are gymnosperms: Cycadophyta, Ginkgophyta, Gnetophyta, and Coniferophyta. It is uncertain how the four phyla of gymnosperms are related to each other. Figure 30.7 surveys the diversity of extant gymnosperms.

Phylum Cycadophyta

The 350 species of living cycads have large cones and palmlike leaves (true palm species are angiosperms). Unlike most seed plants, cycads have flagellated sperm, indicating their descent from seedless vascular plants that had motile sperm. Cycads thrived during the Mesozoic era, known as the age of cycads as well as the age of dinosaurs. Today, however, cycads are the most endangered of all plant groups: 75% of their species are threatened by habitat destruction and other human actions.



Cycas revoluta

Phylum Ginkgophyta



Ginkgo biloba is the only surviving species of this phylum; like cycads, ginkgos have flagellated sperm. Also known as the maidenhair tree, *Ginkgo biloba* has deciduous fanlike leaves that turn gold in autumn. It is a popular ornamental tree in cities because it tolerates air pollution well. Landscapers often plant only pollen-producing trees because the fleshy seeds smell rancid as they decay.

Phylum Gnetophyta

Phylum Gnetophyta includes plants in three genera: *Gnetum*, *Ephedra*, and *Welwitschia*. Some species are tropical, whereas others live in deserts. Although very different in appearance, the genera are grouped together based on molecular data.

► **Welwitschia** (also below). This genus consists of one species, *Welwitschia mirabilis*, a plant that can live for thousands of years and is found only in the deserts of southwestern Africa. Its straplike leaves are among the largest leaves known.



Ovulate cones

► **Ephedra**. This genus includes about 40 species that inhabit arid regions worldwide. These desert shrubs, commonly called “Mormon tea,” produce the compound ephedrine, which is used medicinally as a decongestant.



► **Gnetum**. This genus includes about 35 species of tropical trees, shrubs, and vines, mainly native to Africa and Asia. Their leaves look similar to those of flowering plants, and their seeds look somewhat like fruits.



Phylum Coniferophyta

Phylum Coniferophyta, the largest gymnosperm phyla, consists of about 600 species of conifers (from the Latin *conus*, cone, and *ferre*, to carry), including many large trees. Most species have woody cones, but a few have fleshy cones. Some, such as pines, have needlelike leaves. Others, such as redwoods, have scalelike leaves. Some species dominate vast northern forests, whereas others are native to the Southern Hemisphere.

► **Douglas fir.** This evergreen tree (*Pseudotsuga menziesii*) provides more timber than any other North American tree species. Some uses include house framing, plywood, pulpwood for paper, railroad ties, and boxes and crates.



◄ **European larch.** The needlelike leaves of this deciduous conifer (*Larix decidua*) turn yellow before they are shed in autumn. Native to the mountains of central Europe, including Switzerland's Matterhorn, depicted here, this species is extremely cold-tolerant, able to survive winter temperatures that plunge to -50°C .



► **Sequoia.** This giant sequoia (*Sequoiadendron giganteum*) in California's Sequoia National Park weighs about 2,500 metric tons, equivalent to about 24 blue whales (the largest animals) or 40,000 people. The giant sequoia is one of the largest living organisms and also among the most ancient, with some individuals estimated to be between 1,800 and 2,700 years old. Their cousins, the coast redwoods (*Sequoia sempervirens*), grow to heights of more than 110 m (taller than the Statue of Liberty) and are found only in a narrow coastal strip of northern California and southern Oregon.

Most conifers are evergreens; they retain their leaves throughout the year. Even during winter, a limited amount of photosynthesis occurs on sunny days. When spring comes, conifers already have fully developed leaves that can take advantage of the sunnier, warmer days. Some conifers, such as the dawn redwood, tamarack, and larch, are deciduous trees that lose leaves each autumn.

► **Common juniper.** The "berries" of the common juniper (*Juniperus communis*) are actually ovule-producing cones consisting of fleshy sporophylls.



◄ **Wollemi pine.** Survivors of a conifer group once known only from fossils, living Wollemi pines (*Wollemia nobilis*) were discovered in 1994 in a national park near Sydney, Australia. At that time, the species consisted of 40 known trees. As a result of conservation efforts, it is now widely propagated. The inset photo compares the leaves of this "living fossil" with actual fossils.



► **Bristlecone pine.** This species (*Pinus longaeva*), which is found in the White Mountains of California, includes some of the world's oldest living organisms. One bristlecone pine was recently found to be more than 5,000 years old, and some may be even older.



CONCEPT CHECK 30.2

1. Explain how the pine life cycle in Figure 30.4 reflects the five adaptations common to all seed plants.
2. **VISUAL SKILLS** Based on Figure 30.4, compare and contrast the function of pollination and fertilization in sexual reproduction in pines.
3. **MAKE CONNECTIONS** Early seed plants in genus *Elkinsia* are a sister group to a clade consisting of gymnosperms and angiosperms. Draw a phylogenetic tree of seed plants that shows *Elkinsia*, gymnosperms, and angiosperms; date the branch points on this tree using fossil evidence. (See Figure 26.5.)

For suggested answers, see Appendix A.

CONCEPT 30.3

The reproductive adaptations of angiosperms include flowers and fruits



Nonvascular plants (bryophytes)
Seedless vascular plants
Gymnosperms
Angiosperms

Commonly known as flowering plants, angiosperms are seed plants with the reproductive structures

called flowers and fruits. The name *angiosperm* (from the Greek *angion*, container) refers to seeds contained in fruits. Angiosperms are the most diverse and widespread of all plants, with more than 290,000 species (about 90% of all plant species).

Characteristics of Angiosperms

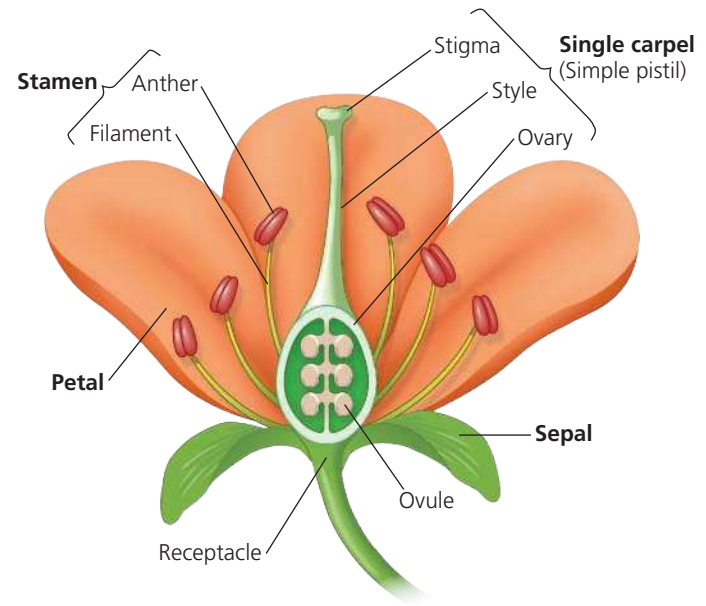
All angiosperms are classified in a single phylum, Anthophyta. Before considering the evolution of angiosperms, we will examine two of their key adaptations—flowers and fruits—and the roles of these structures in the angiosperm life cycle.

Flowers

The **flower** is a unique angiosperm structure that is specialized for sexual reproduction. In many angiosperm species, insects or other animals transfer pollen from one flower to the sex organs on another flower, which makes pollination more directed than the wind-dependent pollination of most species of gymnosperms. However, some angiosperms are wind-pollinated, particularly those species that occur in dense populations, such as grasses and tree species in temperate forests.

A flower is a specialized shoot that can have up to four types of modified leaves called floral organs: sepals, petals, stamens, and carpels (Figure 30.8). Starting at the base of the flower are the **sepals**, which are usually green and enclose the flower before it opens (think of a rosebud). Interior to the sepals are the **petals**, which are brightly colored in most flowers and can aid in attracting pollinators. Flowers that are wind-pollinated, such as grasses, generally lack brightly colored parts. In all angiosperms, the sepals and petals are sterile floral organs, meaning that they do not produce sperm or eggs.

▼ Figure 30.8 The structure of an idealized flower.



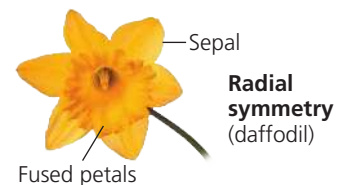
Within the petals are two types of fertile floral organs that produce spores, the stamens and carpels. Stamens and carpels are sporophylls, modified leaves that are specialized for reproduction. **Stamens** are microsporophylls: They produce microspores that develop into pollen grains containing male gametophytes. A stamen consists of a stalk called the **filament** and a terminal sac, the **anther**, where pollen is produced.

Carpels are megasporophylls: They produce megaspores that give rise to female gametophytes. The carpel is the “container” mentioned earlier in which seeds are enclosed; as such, it is a key structure that distinguishes angiosperms from gymnosperms. At the tip of the carpel is a sticky **stigma** that receives pollen. A **style** leads from the stigma to a structure at the base of the carpel, the **ovary**; the ovary contains one or more ovules. As in gymnosperms, each angiosperm ovule contains a female gametophyte. If fertilized, an ovule develops into a seed.

A flower may have one or more carpels. In many species, multiple carpels are fused into one structure. The term **pistil** is sometimes used to refer to a single carpel (a simple pistil) or two or more fused carpels (a compound pistil). Flowers also vary in symmetry (Figure 30.9) and other aspects of shape, as

▼ Figure 30.9 Flower symmetry.

In radial symmetry, the sepals, petals, stamens, and carpels radiate out from a center. Any line through the central axis divides the flower into two equal parts.



In bilateral symmetry, the flower can only be divided into two equal parts by a single line.



DRAW IT Draw the single line that can divide the bilaterally symmetrical flower into two equal parts.

well as size, color, and odor. Much of this diversity results from adaptation to specific pollinators (see Figures 38.4 and 38.5).

Fruits

As seeds develop from ovules after fertilization, the ovary wall thickens and the ovary matures into a **fruit**. A pea pod is an example of a fruit, with seeds (mature ovules, the peas) encased in the ripened ovary (the pod).

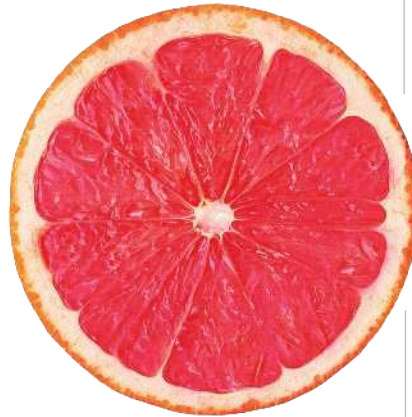
Fruits protect seeds and aid in their dispersal. Mature fruits can be either fleshy or dry (**Figure 30.10**). Tomatoes, plums, and grapes are examples of fleshy fruits, in which the wall (pericarp) of the ovary becomes soft during ripening. Dry fruits include beans, nuts, and grains. Some dry fruits split open at maturity to release seeds, whereas others remain closed. The dry, wind-dispersed fruits of grasses, harvested while on the plant, are major staple foods for humans. The cereal grains of maize, rice, wheat, and other grasses, though easily mistaken for seeds, are each actually a fruit with a dry outer covering (the former wall of the ovary) that adheres to the seed coat of the seed within.

▼ **Figure 30.10** Some variations in fruit structure.

- ▼ Tomato, a fleshy fruit with soft outer and inner layers of pericarp (fruit wall)



- ▼ Ruby grapefruit, a fleshy fruit with a firm outer layer and soft inner layer of pericarp



- ▼ Nectarine, a fleshy fruit with a soft outer layer and hard inner layer (pit) of pericarp



- ▼ Hazelnut, a dry fruit that remains closed at maturity



- ▼ Milkweed, a dry fruit that splits open at maturity



As shown in **Figure 30.11**, various adaptations of fruits and seeds help to disperse seeds (see also Figure 38.12). The seeds of some flowering plants, such as dandelions and maples, are contained within fruits that function like parachutes or propellers, adaptations that enhance dispersal by wind. Some fruits, such as coconuts, are adapted to dispersal by water. And the seeds of many angiosperms are carried by animals. Some angiosperms have fruits modified as burrs that cling to animal fur (or the clothes of humans). Others produce edible fruits, which are usually nutritious, sweet tasting, and vividly colored, advertising their ripeness. When an animal eats the fruit, it digests the fruit's fleshy part, but the tough seeds usually pass unharmed through the animal's digestive tract. When the animal defecates, it may deposit the seeds, along with a supply of natural fertilizer, many kilometers from where the fruit was eaten.

▼ **Figure 30.11** Fruit adaptations that enhance seed dispersal.



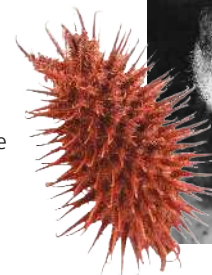
- Some plants have mechanisms that disperse seeds by explosive action.

- Wings enable maple fruits to be carried by the wind.



- Seeds within berries and other edible fruits are often dispersed in animal feces.

- The barbs of cockleburs facilitate seed dispersal by allowing the fruits to "hitchhike" on animals.



➔ **Mastering Biology** Animation: Fruit Structure and Seed Dispersal

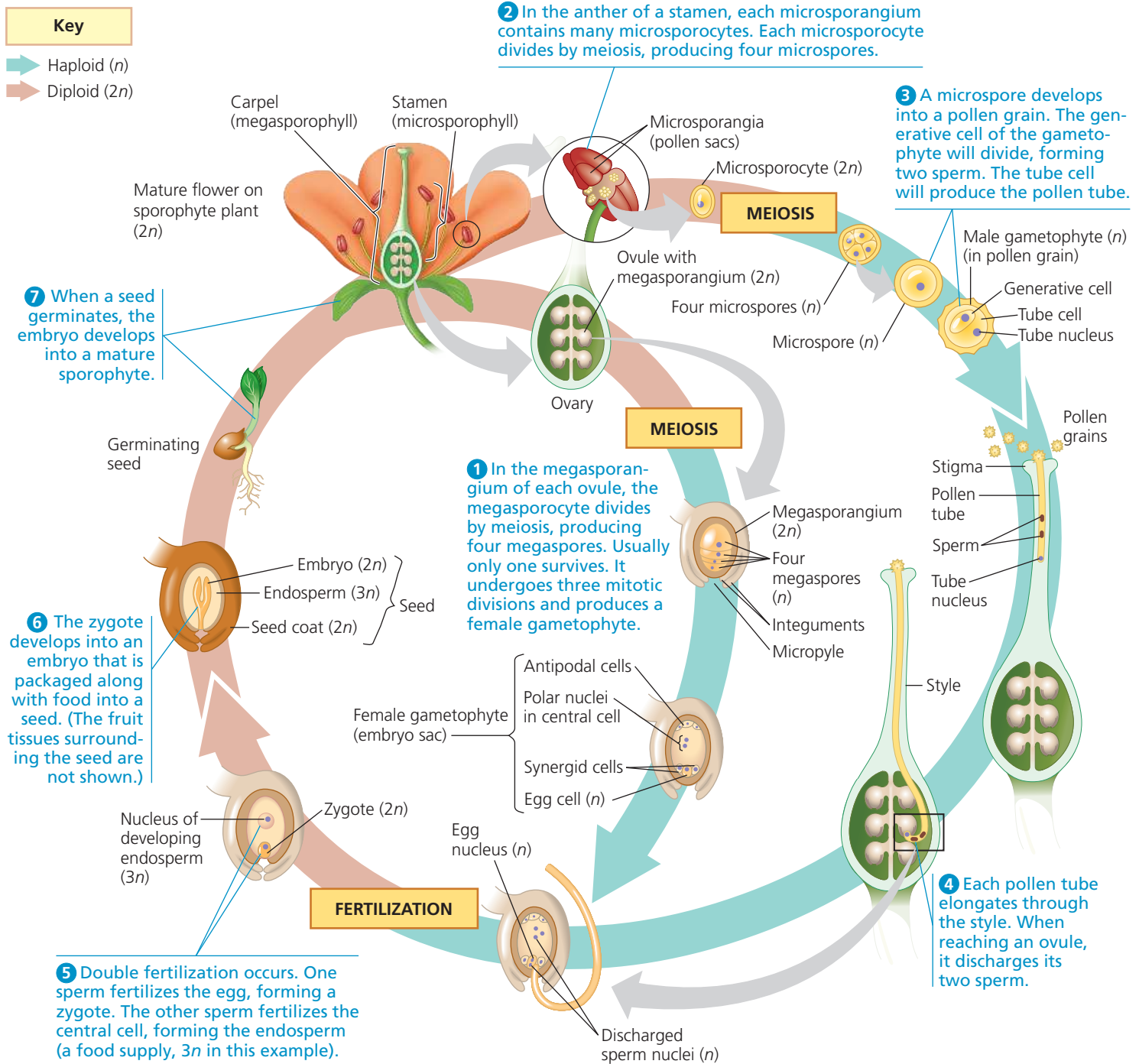
The Angiosperm Life Cycle

You can follow a typical angiosperm life cycle in **Figure 30.12**. The flower of the sporophyte produces microspores that form male gametophytes and megaspores that form female gametophytes. The male gametophytes are in the pollen grains, which develop within microsporangia in the anthers. Each male gametophyte has two haploid cells: a *generative cell* that divides, forming two sperm, and a *tube cell* that produces a

pollen tube. Each ovule, which develops in the ovary, contains a female gametophyte, also known as an **embryo sac**. The embryo sac consists of only a few cells, one of which is the egg.

After its release from the anther, the pollen is carried to the sticky stigma at the tip of a carpel. Although some flowers self-pollinate, most have mechanisms that ensure **cross-pollination**, which in angiosperms is the transfer of pollen from an anther of a flower on one plant to the stigma of a

▼ **Figure 30.12** The life cycle of an angiosperm.



VISUAL SKILLS Based on this figure, what is the maximum number of seeds this flower could produce? To produce that number of seeds, at least how many pollen grains would have to germinate?

➡ **Mastering Biology Figure Walkthrough Animation: Angiosperm Life Cycle**

flower on another plant of the same species. Cross-pollination enhances genetic variability. In some species, stamens and carpels of a single flower may mature at different times, or they may be so arranged that self-pollination is unlikely.

The pollen grain absorbs water and germinates after it adheres to the stigma of a carpel. The tube cell produces a pollen tube that grows down within the style of the carpel. After reaching the ovary, the pollen tube penetrates through the **micropyle**, a pore in the integuments of the ovule, and discharges two sperm cells into the female gametophyte (embryo sac). One sperm fertilizes the egg, forming a diploid zygote. The other sperm fuses with the two nuclei in the large central cell of the female gametophyte, producing a triploid cell ($3n$). This type of **double fertilization**, in which one fertilization event produces a zygote and the other produces a triploid cell, is unique to angiosperms.

After double fertilization, the ovule matures into a seed. The zygote develops into a sporophyte embryo with a rudimentary root and one or two seed leaves called **cotyledons**. The triploid central cell of the female gametophyte develops into **endosperm**, tissue rich in starch and other food reserves that nourish the developing embryo.

What is the function of double fertilization in angiosperms? One hypothesis is that double fertilization synchronizes the development of food storage in the seed with the development of the embryo. If a particular flower is not pollinated or sperm cells are not discharged into the embryo sac, fertilization does not occur, and neither endosperm nor embryo forms. So perhaps double fertilization is an adaptation that prevents flowering plants from squandering nutrients on infertile ovules.

Another type of double fertilization occurs in some gymnosperm species belonging to the phylum Gnetophyta. However, double fertilization in these species gives rise to two embryos rather than to an embryo and endosperm.

As you read earlier, the seed consists of the embryo, the endosperm, and a seed coat derived from the integuments. An ovary develops into a fruit as its ovules become seeds. After being dispersed, a seed may germinate if environmental conditions are favorable. The coat ruptures and the embryo emerges as a seedling, using food stored in the endosperm and cotyledons until it can produce its own food by photosynthesis.

Angiosperm Evolution

Charles Darwin once referred to the origin of angiosperms as an “abominable mystery.” He was particularly troubled by the

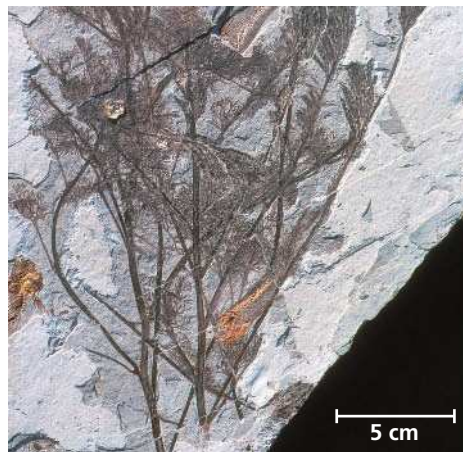
relatively sudden and geographically widespread appearance of angiosperms in the fossil record (about 100 million years ago, based on fossils known to Darwin). Recent fossil evidence and phylogenetic analyses have led to progress in solving Darwin’s mystery, but we still do not fully understand how angiosperms arose from earlier seed plants.

Fossil Angiosperms

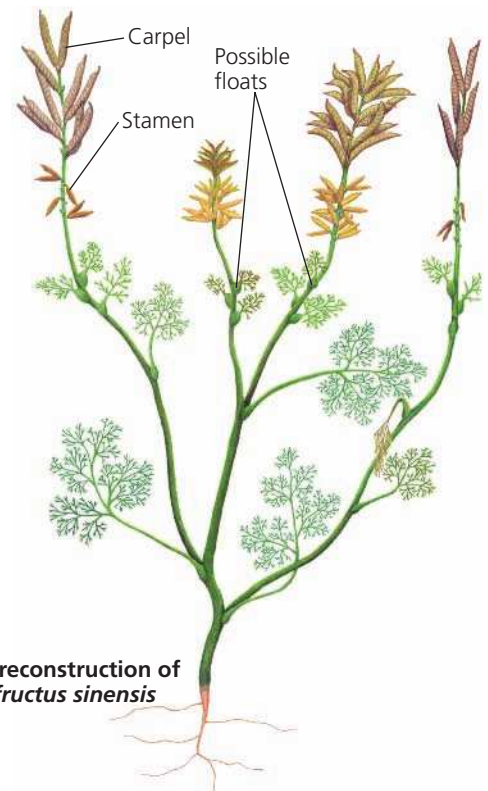
Angiosperms are now thought to have originated in the early Cretaceous period, about 140 million years ago. By the mid-Cretaceous (100 million years ago), angiosperms began to dominate some terrestrial ecosystems. Landscapes changed dramatically as conifers and other gymnosperms gave way to flowering plants in many parts of the world. The Cretaceous ended 66 million years ago with mass extinctions of dinosaurs and many other animal groups and with further increases in the diversity and importance of angiosperms.

What evidence suggests that angiosperms arose 140 million years ago? First, although pollen grains are common in rocks from the Jurassic period (201 to 145 million years ago), none of these pollen fossils have features characteristic of angiosperms, suggesting that angiosperms may have originated after the Jurassic. Indeed, the earliest fossils with distinctive angiosperm features are of 130-million-year-old pollen grains discovered in China, Israel, and England. Early fossils of larger flowering plant structures include those of *Archaeofructus* (Figure 30.13) and *Leefructus*, both of which were discovered in China in rocks that are 125 million years old.

▼ **Figure 30.13** An early flowering plant.



(a) *Archaeofructus sinensis*, a 125-million-year-old fossil. This herbaceous species had simple flowers and bulbous structures that may have served as floats, suggesting it was aquatic. Recent phylogenetic analyses indicate that *Archaeofructus* may belong to the water lily group.



(b) Artist's reconstruction of *Archaeofructus sinensis*

Overall, early angiosperm fossils indicate that the group arose and began to diversify over a 20- to 30-million-year period—a less sudden event than was suggested by the fossils known during Darwin’s lifetime.

Can we infer traits of the angiosperm common ancestor from traits found in early fossil angiosperms? *Archaeofructus*, for example, was herbaceous and had bulbous structures that may have served as floats, suggesting it was aquatic. But investigating whether the angiosperm common ancestor was herbaceous and aquatic also requires examining fossils of other seed plants thought to have been closely related to angiosperms. All of those plants were woody, indicating that the common ancestor was probably woody and probably not aquatic. As we’ll see, this conclusion has been supported by recent phylogenetic analyses.

Angiosperm Phylogeny

Molecular and morphological evidence suggests that extant gymnosperm lineages had diverged from the lineage leading to angiosperms by 305 million years ago. Note that this does not imply that angiosperms originated 305 million years ago, but that the most recent common ancestor of extant gymnosperms and angiosperms lived at that time. Indeed, angiosperms may be more closely related to several extinct lineages of woody seed plants than they are to gymnosperms. One such lineage is the Bennettitales, an extinct group with

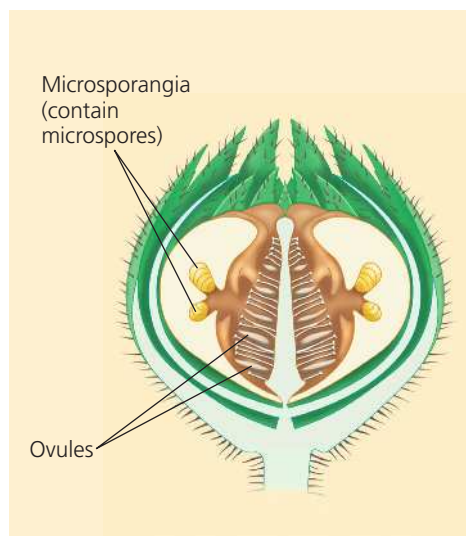
flowerlike structures that may have been pollinated by insects (**Figure 30.14a**). However, the Bennettitales and other similar lineages of extinct woody seed plants did not have carpels or flowers and hence are not classified as angiosperms.

Making sense of the origin of angiosperms also depends on working out the order in which angiosperm clades diverged from one another. Here, dramatic progress has been made in recent years. Molecular and morphological evidence suggests that the shrub *Amborella trichopoda*, water lilies, and star anise are living representatives of lineages that diverged from other angiosperms early in the history of the group (**Figure 30.14b**). *Amborella* is woody, supporting the conclusion mentioned earlier that the angiosperm common ancestor was probably woody. Like the Bennettitales, *Amborella*, water lilies, and star anise lack *vessel elements*, efficient water-conducting cells that are found in most present-day angiosperms. Overall, based on the features of ancestral species and angiosperms like *Amborella*, researchers have hypothesized that early angiosperms were woody shrubs that had small flowers and relatively simple water-conducting cells.

Evolutionary Links with Animals

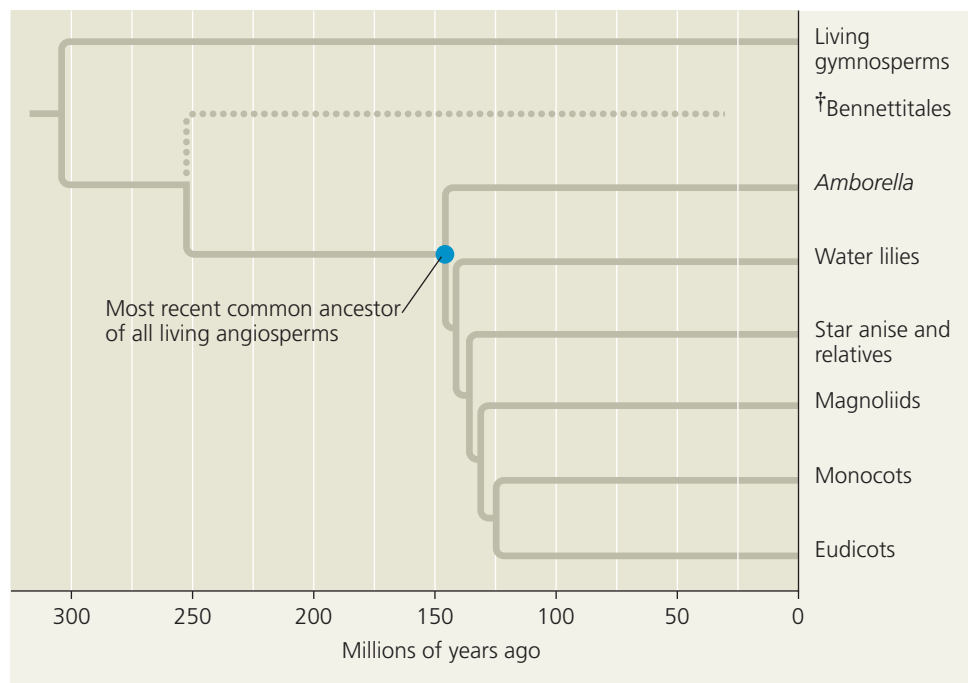
Plants and animals have interacted for hundreds of millions of years, and those interactions have led to evolutionary change. For example, herbivores can reduce a plant’s reproductive success by eating its roots, leaves, or seeds. As a result,

▼ **Figure 30.14** Angiosperm evolutionary history.



(a) A close relative of the angiosperms?

This reconstruction shows a longitudinal section through the flowerlike structures found in the Bennettitales, an extinct group of woody seed plants hypothesized to be more closely related to extant angiosperms than to extant gymnosperms.



(b) Angiosperm phylogeny. This tree represents a current hypothesis of angiosperm evolutionary relationships, based on morphological and molecular evidence. Angiosperms originated about 140 million years ago. The dotted line indicates the uncertain position of the Bennettitales, which may be the sister taxon to the angiosperms.

VISUAL SKILLS Would the branching order of the phylogeny in (b) necessarily have to be redrawn if a 150-million-year-old fossil monocot were discovered? Explain.



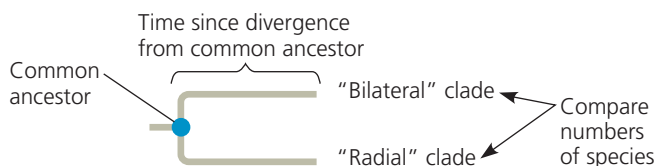
◀ **Figure 30.15 A bee pollinating a bilaterally symmetrical flower.**

To harvest nectar from this Scottish broom flower, a honeybee must land as shown. This releases a tripping mechanism that arches the flower's stamens over the bee and dusts it with pollen. Later, some of this pollen may rub off onto the stigma of the next flower the bee visits.

➔ **Mastering Biology Video: Bee Pollinating**

if an effective defense against herbivores originates in a group of plants, those plants may be favored by natural selection—as will herbivores that overcome this new defense. Plant-pollinator and other mutually beneficial interactions also can have such reciprocal evolutionary effects.

Plant-pollinator interactions also may have affected the rates at which new species form. Consider the impact of a flower's symmetry (see Figure 30.9). On a flower with bilateral symmetry, an insect pollinator can obtain nectar (a sugary solution secreted by flower glands) only when landing in a certain position (**Figure 30.15**). This constraint makes it more likely that pollen is placed on a part of the insect's body that will come into contact with the stigma of a flower of the same species. Such specificity of pollen transfer reduces gene flow between diverging populations and could lead to increased rates of speciation in plants with bilateral symmetry. This hypothesis can be tested using the approach illustrated in this diagram:



A key step in this approach is to identify cases in which a clade with bilaterally symmetric flowers shares an immediate common ancestor with a clade whose members have radially symmetric flowers. One recent study identified 19 pairs of closely related “bilateral” and “radial” clades. On average, the clade with bilaterally symmetric flowers had nearly 2,400 more species than did the related clade with radial symmetry. This result suggests that flower shape can affect the rate at which new species form, perhaps by affecting the behavior of insect pollinators. Overall, plant-pollinator interactions may have contributed to the increasing dominance of flowering plants in the Cretaceous period, making angiosperms centrally important to ecological communities.

Angiosperm Diversity

From their humble beginnings in the Cretaceous period, angiosperms have diversified into more than 290,000 living species. Until the late 1990s, most systematists divided flowering plants into two groups, based partly on the number of cotyledons, or seed leaves, in the embryo. Species with one cotyledon were called **monocots**, and those with two were called **dicots**. Other features, such as flower and leaf structure, were also used to define the two groups. Recent DNA studies, however, indicate that the species traditionally called dicots are paraphyletic. The vast majority of species once categorized as dicots form a large clade, now known as **eudicots** (“true” dicots). **Figure 30.16** compares the main characteristics of monocots and eudicots. The rest of the former dicots are now grouped into four small lineages. Three of these lineages—*Amborella*, water lilies, and star anise and relatives—are informally called **basal angiosperms** because they diverged from other angiosperms early in the history of the group (see Figure 30.14b). A fourth lineage, the **magnoliids**, evolved later.

Figure 30.17 provides an overview of angiosperm diversity.

▼ **Figure 30.16 Characteristics of monocots and eudicots.**

	Embryos	Leaf venation	Stems	Roots	Pollen	Flowers
Monocot Characteristics	 One cotyledon	 Veins usually parallel	 Vascular tissue scattered	 Root system usually fibrous (no main root)	 Pollen grain with one opening	 Floral organs usually in multiples of three
Eudicot Characteristics	 Two cotyledons	 Veins usually netlike	 Vascular tissue usually arranged in ring	 Taproot (main root) usually present	 Pollen grain with three openings	 Floral organs usually in multiples of four or five

Basal Angiosperms

Surviving basal angiosperms consist of three lineages comprising only about 100 species. The first lineage to have diverged from other angiosperms is represented today by a single species, *Amborella trichopoda* (right). The other surviving lineages diverged later: a clade that includes water lilies and a clade consisting of the star anise and its relatives.



◀ **Water lily (*Nymphaea* "Rene Gerard").** Species of water lilies are found in aquatic habitats throughout the world. Water lilies belong to a clade that diverged from other angiosperms early in the group's history.



▶ ***Amborella trichopoda*.** This small shrub, found only on the South Pacific island of New Caledonia, may be the sole survivor of a branch at the base of the angiosperm tree.

▶ **Star anise (*Illicium*).** This genus belongs to a third surviving lineage of basal angiosperms.



Magnoliids

Magnoliids consist of about 8,500 species, most notably magnolias, laurels, and black pepper plants. They include both woody and herbaceous species. Although they share some traits with basal angiosperms, such as a typically spiral rather than whorled arrangement of floral organs, magnoliids are more closely related to eudicots and monocots.



▶ **Southern magnolia (*Magnolia grandiflora*).** This member of the magnolia family is a large tree. The variety of southern magnolia shown here, called "Goliath," has flowers that measure up to about a foot across.

Monocots

About one-quarter of angiosperm species are monocots —about 72,000 species. Some of the largest groups are the orchids, grasses, and palms. Grasses include some of the most agriculturally important crops, such as maize, rice, and wheat.



▶ **Orchid (*Paphiopedilum callosum*)**

▶ **Barley (*Hordeum vulgare*), a grass**



▶ **Pygmy date palm (*Phoenix roebelenii*)**

Eudicots

More than two-thirds of angiosperm species are eudicots—roughly 210,000 species. The largest group is the legume family, which includes such crops as peas and beans. Also important economically is the rose family, which includes many plants with ornamental flowers as well as some species with edible fruits, such as strawberry plants and apple and pear trees. Most of the familiar flowering trees are eudicots, such as oak, walnut, maple, willow, and birch.



▶ **Snow pea (*Pisum sativum*), a legume**

▶ **Dog rose (*Rosa canina*), a wild rose**



▶ **Armenian oak (*Quercus pontica*)**

CONCEPT CHECK 30.3

1. It is said that an oak is an acorn's way of making more acorns. Write an explanation that includes these terms: sporophyte, gametophyte, ovule, seed, ovary, and fruit.
2. Compare and contrast a pine cone and a flower in terms of structure and function.
3. **WHAT IF?** Do speciation rates in closely related clades of flowering plants show that flower shape is *correlated with* the rate at which new species form or that flower shape is *responsible for* this rate? Explain.

For suggested answers, see Appendix A.

CONCEPT 30.4

Human welfare depends on seed plants

In forests and on farms, seed plants are key sources of food, fuel, wood products, and medicine. Our reliance on them makes the preservation of plant diversity critical.

Products from Seed Plants

Most of our food comes from angiosperms. Just six crops—maize, rice, wheat, potatoes, cassava, and sweet potatoes—yield 80% of all the calories consumed by humans. We also depend on angiosperms to feed livestock: It takes 5–7 kg of grain to produce 1 kg of grain-fed beef.

Today's crops are the products of artificial selection—the result of plant domestication that began about 12,000 years ago. To appreciate the scale of this transformation, note how the number and size of seeds in domesticated plants are greater than those of their wild relatives, as in the case of maize and the grass teosinte (see Figure 38.16). Scientists can glean information about domestication by comparing the genes of crops with those of wild relatives. With maize, dramatic changes such as increased cob size and loss of the hard coating around teosinte kernels may have been initiated by as few as five mutations.

Flowering plants also provide other edible products. Two popular beverages come from tea leaves and coffee beans, and you can thank the cacao tree for cocoa and chocolate. Spices are derived from various plant parts, such as flowers (cloves, saffron), fruits and seeds (vanilla, black pepper, mustard), leaves (basil, mint, sage), and even bark (cinnamon).

Many seed plants are sources of wood, which is absent in all living seedless plants. Wood consists of tough-walled xylem cells (see Figure 35.22). It is the primary source of fuel for much of the world, and wood pulp, typically derived from conifers such as fir and pine, is used to make paper. Wood remains the most widely used construction material.

For centuries, humans have also depended on seed plants for medicines. Many cultures use herbal remedies, and scientists have extracted and identified medicinally active compounds from many of these plants and later synthesized

Table 30.1 Examples of Plant-Derived Medicines

Compound	Source	Use
Atropine	Belladonna plant	Eye pupil dilator
Digitalin	Foxglove	Heart medication
Menthol	Eucalyptus tree	Throat soother
Quinine	Cinchona tree	Malaria preventive
Taxol	Pacific yew	Ovarian cancer drug
Tubocurarine	Curare tree	Muscle relaxant
Vinblastine	Periwinkle	Leukemia drug

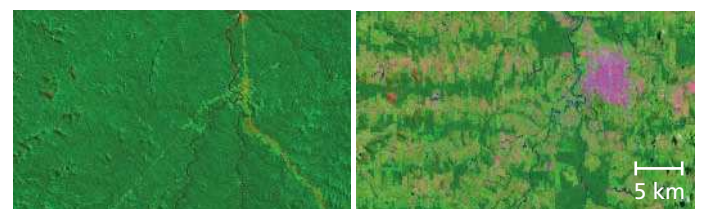
them. Willow leaves and bark have long been used in pain-relieving remedies, including prescriptions by the Greek physician Hippocrates. In the 1800s, scientists traced the willow's medicinal property to the chemical salicin. A synthesized derivative, acetylsalicylic acid, is what we call aspirin. Plants are also a direct source of medicinal compounds (**Table 30.1**). In the United States, about 25% of prescription drugs contain an active ingredient from plants, usually seed plants.

Threats to Plant Diversity

Although plants may be a renewable resource, plant diversity is not. The exploding human population and its demand for space and resources are threatening plant species across the globe. The problem is especially severe in the tropics, where more than two-thirds of the human population live and where population growth is fastest. About 63,000 km² (15 million acres) of tropical rain forest are cleared each year (**Figure 30.18**), a rate that would completely eliminate the remaining 11 million km² of tropical forests in 175 years. The loss of forests reduces the absorption of atmospheric carbon dioxide (CO₂) that occurs during photosynthesis, potentially contributing to global warming. Also, as forests disappear, so do large numbers of plant species. Of course, once a species becomes extinct, it can never return.

The loss of plant species is often accompanied by the loss of insects and other rain forest animals. Scientists estimate that if current rates of loss in the tropics and elsewhere

▼ **Figure 30.18** **Clear-cutting of tropical forests.** Over the past several hundred years, nearly half of Earth's tropical forests have been cut down and converted to farmland and other uses. A satellite image from 1975 (left) shows a dense forest in Brazil. By 2012, much of this forest had been cut down. Deforested and urban areas are shown as light purple.



continue, 50% or more of Earth's species will become extinct within the next few centuries. Such losses would constitute a global mass extinction, rivaling the Permian and Cretaceous mass extinctions and forever changing the evolutionary history of plants (and many other organisms).

Many people have ethical concerns about contributing to the extinction of species. In addition, there are practical reasons to be concerned about the loss of plant diversity. So far, we have explored the potential uses of only a tiny fraction of the more than 325,000 known plant species. For example, almost all our food is based on the cultivation of only about two dozen species of seed plants. And fewer than 5,000 plant species have been studied as potential sources of medicines.

The tropical rain forest may be a medicine chest of healing plants that could be extinct before we even know they exist. If we begin to view rain forests and other ecosystems as living treasures that can regenerate only slowly, we may learn to harvest their products at sustainable rates.

CONCEPT CHECK 30.4

1. Explain why plant diversity can be considered a nonrenewable resource.
2. **WHAT IF?** How could phylogenies be used to help researchers search more efficiently for novel medicines derived from seed plants?

For suggested answers, see Appendix A.

30 Chapter Review

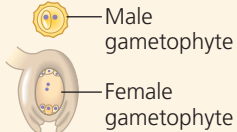
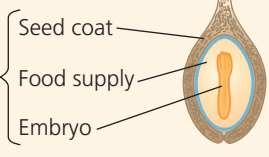
➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zk9t.

CONCEPT 30.1

Seeds and pollen grains are key adaptations for life on land (pp. 637–639)

Five Derived Traits of Seed Plants	
Reduced gametophytes	Microscopic male and female gametophytes (n) are nourished and protected by the sporophyte ($2n$) 
Heterospory	Microspore (gives rise to a male gametophyte)  Megaspore (gives rise to a female gametophyte) 
Ovules	Ovule (gymnosperm) Integument ($2n$) Megaspore (n) Megasporangium ($2n$) 
Pollen	Pollen grains make water unnecessary for fertilization 
Seeds	Seeds: survive better than unprotected spores, can be transported long distances 

? Describe how the parts of an ovule (integument, megaspore, megasporangium) correspond to the parts of a seed.

CONCEPT 30.2

Gymnosperms bear “naked” seeds, typically on cones (pp. 640–644)

- Dominance of the sporophyte generation, the development of seeds from fertilized ovules, and the role of pollen in transferring sperm to ovules are key features of a typical gymnosperm life cycle.
- Gymnosperms appear early in the plant fossil record and dominated many Mesozoic terrestrial ecosystems. Living seed plants can be divided into two monophyletic groups: gymnosperms and angiosperms. Extant gymnosperms include cycads, *Ginkgo biloba*, gnetophytes, and **conifers**.

? Although there are just over 1,000 species of gymnosperms, the group is still very successful in terms of its evolutionary longevity, adaptations, and geographic distribution. Explain.

CONCEPT 30.3

The reproductive adaptations of angiosperms include flowers and fruits (pp. 644–651)

- **Flowers** generally consist of four types of modified leaves: **sepals**, **petals**, **stamens** (which produce pollen), and **carpels** (which produce ovules). **Ovaries** ripen into **fruits**, which often carry seeds by wind, water, or animals to new locations.
- Flowering plants originated about 140 million years ago, and by the mid-Cretaceous (100 mya) had begun to dominate some terrestrial ecosystems. Fossils and phylogenetic analyses offer insights into the origin of flowers.
- Several groups of **basal angiosperms** have been identified. Other major clades of angiosperms include **magnoliids**, **monocots**, and **eudicots**.
- Pollination and other interactions between angiosperms and animals may have contributed to the success of flowering plants during the last 100 million years.

? Explain why Darwin called the origin of angiosperms an “abominable mystery,” and describe what has been learned from fossil evidence and phylogenetic analyses.

CONCEPT 30.4

Human welfare depends on seed plants (pp. 651–652)

- Humans depend on seed plants for products such as food, wood, and many medicines.
- Destruction of habitat threatens the extinction of many plant species and the animal species they support.

? Explain why destroying the remaining tropical forests might harm humans and lead to a mass extinction.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Where in an angiosperm would you find a megasporangium?
 - in the style of a flower
 - enclosed in the stigma of a flower
 - within an ovule contained within an ovary of a flower
 - packed into pollen sacs within the anthers found on a stamen
- Which of the following is a key feature of seed plants facilitating life on land?
 - homospory
 - pollen
 - reduced sporophytes
 - spores
- In angiosperms, which of the following is correctly paired with its chromosome count?

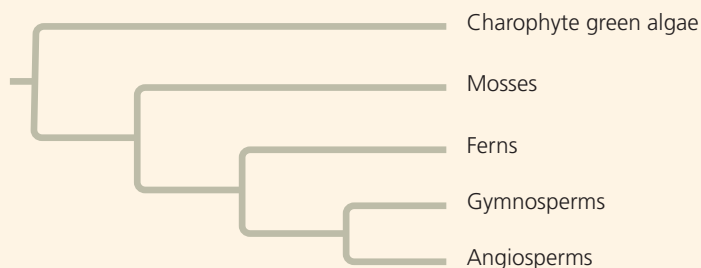
(A) microspore— n	(C) egg— $2n$
(B) zygote— n	(D) megaspore— $2n$
- Which of the following is a characteristic that distinguishes gymnosperms and angiosperms from other plants?
 - alternation of generations
 - independent gametophytes
 - vascular tissue
 - ovules
- Which of the following are present in angiosperms but not in gymnosperms?

(A) seeds	(C) ovaries
(B) pollen	(D) ovules

Levels 3-4: Applying/Analyzing

- DRAW IT** Use the letters a–d to label where on the phylogenetic tree each of the following derived characters appears.

a. flowers	c. seeds
b. embryos	d. vascular tissue



- EVOLUTION CONNECTION** The history of life has been punctuated by several mass extinctions. For example, the impact of a meteorite may have wiped out most of the dinosaurs and many forms of marine life at the end of the Cretaceous period (see Concept 25.4). Fossils indicate that plants were less severely affected by this mass extinction. What adaptations may have enabled plants to withstand this disaster better than animals?

Levels 5-6: Evaluating/Creating

- SCIENTIFIC INQUIRY • DRAW IT** As will be described in detail in Concept 38.1, the female gametophyte of angiosperms typically has seven cells, one of which, the central cell, contains two haploid nuclei. After double fertilization, the central cell develops into endosperm, which is triploid. Because magnoliids, monocots, and eudicots typically have female gametophytes with seven cells and triploid endosperm, scientists assumed that this was the ancestral state for angiosperms. Consider, however, the following recent discoveries:
 - Our understanding of angiosperm phylogeny has changed to that shown in Figure 30.14b.
 - *Amborella trichopoda* has eight-celled female gametophytes and triploid endosperm.
 - Water lilies and star anise have four-celled female gametophytes and diploid endosperm.
 - Draw a phylogeny of the angiosperms (see Figure 30.14b), incorporating the data given above about the number of cells in female gametophytes and the ploidy of the endosperm. Assume that all of the star anise relatives have four-celled female gametophytes and diploid endosperm.
 - What does your labeled phylogeny suggest about the evolution of the female gametophyte and endosperm in angiosperms?
- WRITE ABOUT A THEME: ORGANIZATION** Cells are the basic units of structure and function in all organisms. A key feature in the life cycle of plants is the alternation of multicellular haploid and diploid generations. Imagine a lineage of flowering plants in which mitotic cell division did not occur between the events of meiosis and fertilization (see Figure 30.12). In a short essay (100–150 words), describe how this change in the timing of cell division would affect the structure and life cycle of plants in this lineage.

10. SYNTHESIZE YOUR KNOWLEDGE

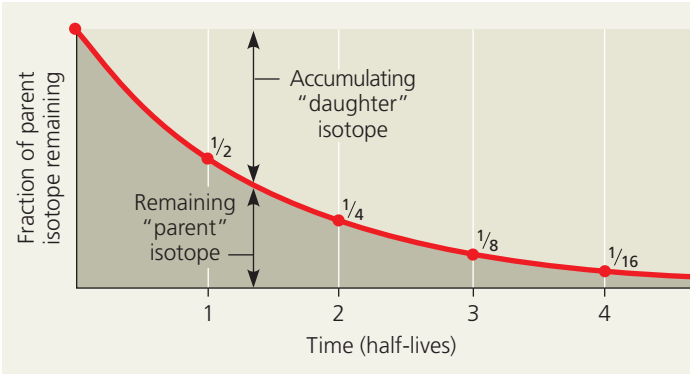


This photograph shows a milkweed seed in flight. Describe how seeds and other adaptations in seed plants contributed to the rise of seed plants and their dominant role in plant communities today.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom AAAS
 How have Earth's forests changed over time?
 Go to "Seeing the Forest for the Trees"
 at www.scienceintheclassroom.org.
 ➔ **Instructors:** Questions can be assigned in Mastering Biology.

▼ **Figure 25.6 Radiometric dating.** In this diagram, each unit of time represents one half-life of a radioactive isotope.



DRAW IT Relabel the x-axis of this graph in years to illustrate the radioactive decay of uranium-238 (half-life = 4.5 billion years).

isotope to decay (Figure 25.6). Each type of radioactive isotope has a characteristic half-life, which is not affected by temperature, pressure, or other environmental variables. For example, carbon-14 decays relatively quickly; its half-life is 5,730 years. Uranium-238 decays slowly; its half-life is 4.5 billion years.

Fossils contain isotopes of elements that accumulated in the organisms when they were alive. For example, a living organism contains the most common carbon isotope, carbon-12, as well as a radioactive isotope, carbon-14. When the organism dies, it stops accumulating carbon, and the amount of carbon-12 in its tissues does not change over time. However, the carbon-14 that it contains at the time of death slowly decays into another element, nitrogen-14. Thus, by measuring the ratio of carbon-14 to carbon-12 in a fossil, we can determine the fossil’s age. This method works for fossils up to about 75,000 years old; fossils older than that contain too little carbon-14 to be detected with current techniques. Radioactive isotopes with longer half-lives are used to date older fossils.

Determining the age of these older fossils in sedimentary rocks can be challenging. Organisms do not use radioisotopes with long half-lives, such as uranium-238, to build their bones or shells. In addition, sedimentary rocks are often composed of sediments of differing ages. Although we cannot date these older fossils directly, an indirect method can be used to infer the age of fossils that are sandwiched between two layers of volcanic rock. As lava cools into volcanic rock, radioisotopes from the surrounding environment become trapped in the newly formed rock. Some of the trapped radioisotopes have long half-lives, allowing geologists to estimate the ages of ancient volcanic rocks. If two volcanic layers surrounding fossils are found to be 525 million and 535 million years old, for example, then the fossils are roughly 530 million years old.

The study of fossils has helped geologists establish a geologic record: a standard time scale that divides Earth’s history into four eons and further subdivisions (Table 25.1). We’ll describe key events that occurred during these different time periods in Concept 25.3.

Table 25.1 The Geologic Record						
Eon	Era	Period	Epoch	Age (Millions of Years Ago)		
	Cenozoic	Quaternary	Holocene	0.01		
			Pleistocene	2.6		
		Neogene	Pliocene	5.3		
			Miocene	23		
		Paleogene	Oligocene	33.9		
			Eocene	56		
			Paleocene	66		
			Phan-ero-zoic	Mesozoic	Cretaceous	
		Jurassic			201	
		Triassic			252	
	Paleozoic	Permian		299		
		Carboniferous		359		
		Devonian		419		
		Silurian		444		
		Ordovician		485		
		Cambrian		541		
		Protero-zoic	Neo-protero-zoic	Ediacaran		635
				1,000		
				2,500		
Archaean				4,000		
Hadean				Approx. 4,600		

The Origin of New Groups of Organisms

Some fossils document the origin of new groups of organisms. Such fossils are central to our understanding of evolution; they illustrate how new features arise and how long it takes for such changes to occur. We’ll examine one such case here and in Figure 25.7: the origin of mammals.

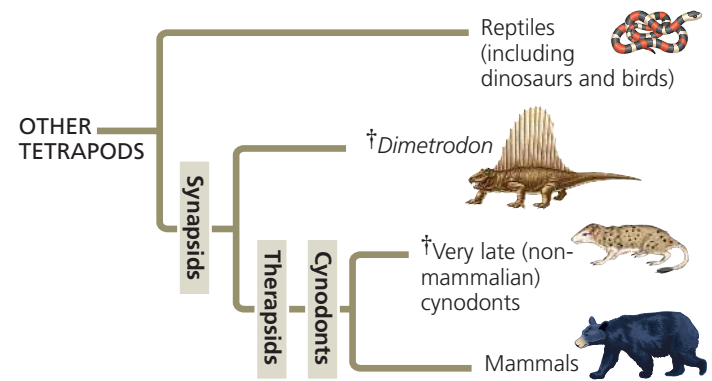
Along with amphibians and reptiles, mammals belong to the group of animals called *tetrapods* (from the Greek *tetra*, four, and *pod*, foot), named for having four limbs. Mammals have a number of unique anatomical features that fossilize readily, allowing scientists to trace their origin. For example, the lower jaw is composed of one bone (the dentary) in mammals but several bones in other tetrapods. In addition, the lower and upper jaws in mammals hinge between a different set of bones than in other tetrapods. Mammals also have a unique set of three bones that transmit sound in the middle ear, the hammer, anvil, and stirrup, whereas other tetrapods have only one such bone, the stirrup (see Concept 34.6). Finally, the teeth of mammals are differentiated into incisors (for tearing), canines (for piercing), and the multi-pointed premolars and molars

▼ Figure 25.7 Exploring the Origin of Mammals

Over the course of 120 million years, mammals originated gradually from a group of tetrapods called synapsids. Shown here are a few of the many fossil organisms whose morphological features represent intermediate steps between living mammals and their early synapsid ancestors. The evolutionary context of the origin of mammals is shown in the tree diagram at right (the dagger symbol † indicates extinct lineages).

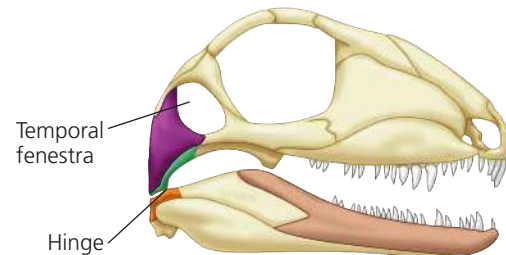
Key to skull bones

Articular	Dentary
Quadrate	Squamosal



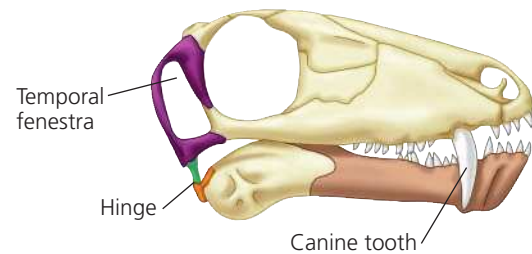
Synapsid (300 mya)

Early synapsids had multiple bones in the lower jaw and single-pointed teeth. The jaw hinge was formed by the articular and quadrate bones. Early synapsids also had an opening called the *temporal fenestra* behind the eye socket. Powerful cheek muscles for closing the jaws probably passed through the temporal fenestra. Over time, this opening enlarged and moved in front of the hinge between the lower and upper jaws, thereby increasing the power and precision with which the jaws could be closed (much as moving a doorknob away from the hinge makes a door easier to close).



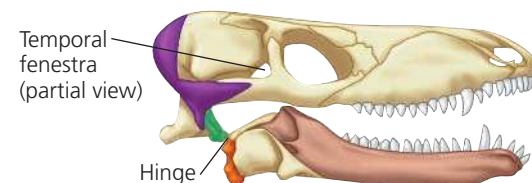
Therapsid (280 mya)

Later, a group of synapsids called therapsids appeared. Therapsids had large dentary bones, long faces, and the first examples of specialized teeth, large canines. These trends continued in a group of therapsids called cynodonts.



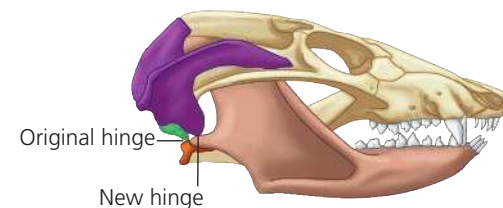
Early cynodont (260 mya)

In early cynodont therapsids, the dentary was the largest bone in the lower jaw, the temporal fenestra was large and positioned forward of the jaw hinge, and teeth with several cusps first appeared (not visible in the diagram). As in earlier synapsids, the jaw had an articular-quadrate hinge.



Later cynodont (220 mya)

Later cynodonts had teeth with complex cusp patterns, and their lower and upper jaws hinged in two locations: They retained the original articular-quadrate hinge and formed a new, second hinge between the dentary and squamosal bones. (The temporal fenestra is not visible in this or the below cynodont skull at the angles shown.)



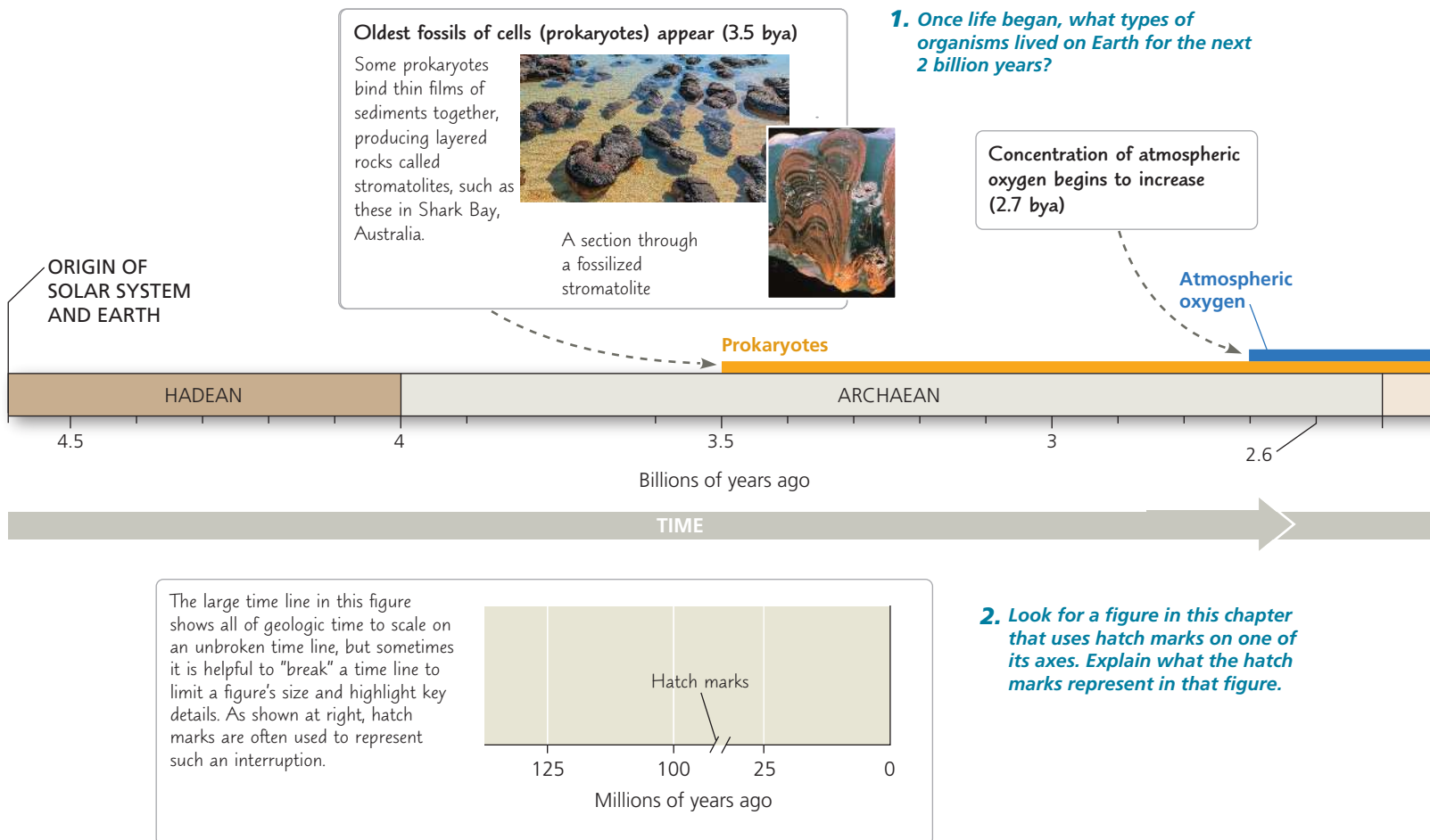
Very late cynodont (195 mya)

In some very late (nonmammalian) cynodonts and early mammals, the original articular-quadrate hinge was lost, leaving the dentary-squamosal hinge as the only hinge between the lower and upper jaws, as in living mammals. The articular and quadrate bones migrated into the ear region (not shown), where they functioned in transmitting sound. In the mammal lineage, these two bones later evolved into the familiar hammer (malleus) and anvil (incus) bones of the ear.



▼ Figure 25.8 VISUALIZING THE SCALE OF GEOLOGIC TIME

Geologic time is so vast that it can be difficult to visualize when key events in the history of life on Earth occurred. In this time line, time runs from left to right, from 4.6 billion years ago to the present. The fossils along the top illustrate representative organisms from different points in time.



(for crushing and grinding). In contrast, the teeth of other tetrapods usually consist of a row of undifferentiated, single-pointed teeth.

As detailed in Figure 25.7, the fossil record shows that the unique features of mammalian jaws and teeth evolved gradually, in a series of steps. Keep in mind that the figure includes just a few examples of the fossil skulls that document the origin of mammals. If all the known fossils in the sequence were arranged by shape and placed side by side, their features would blend smoothly from one group to the next. Some of these fossils would reflect how the features of a group that dominates life today, the mammals, gradually arose in a previously existing group, the cynodonts. Others would reveal side branches on the tree of life—groups of organisms that thrived for millions of years but ultimately left no descendants that survive today.

CONCEPT CHECK 25.2

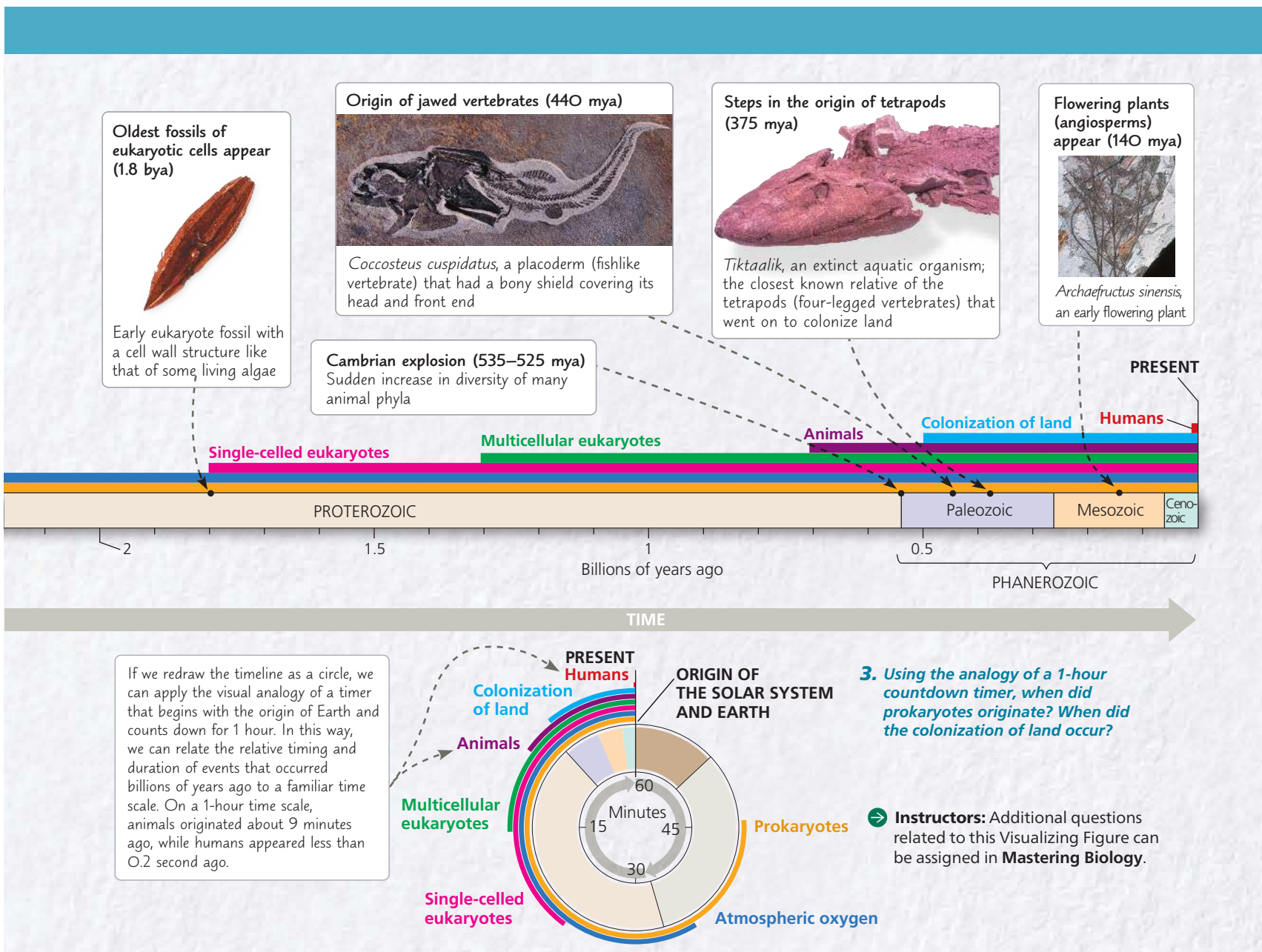
- Describe an example from the fossil record that shows how life has changed over time.
- WHAT IF?** A fossilized skull you unearthed has a carbon-14/carbon-12 ratio about $\frac{1}{16}$ that of the skulls of present-day animals. Approximately how many years old is the fossilized skull?

For suggested answers, see Appendix A.

CONCEPT 25.3

Key events in life's history include the origins of unicellular and multicellular organisms and the colonization of land

The first three eons in the geologic record—the Hadean, the Archaean, and the Proterozoic—together lasted about



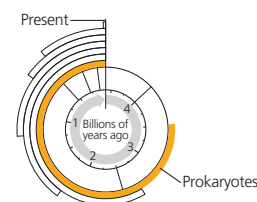
4 billion years. The Phanerozoic eon, roughly the last half billion years, encompasses most of the time that animals have existed on Earth. It is divided into three eras: the Paleozoic, Mesozoic, and Cenozoic. Each era represents a distinct age in the history of Earth and its life. For example, the Mesozoic era is sometimes called the “age of reptiles” because of its abundance of reptilian fossils, including those of dinosaurs. The boundaries between the eras correspond to major extinction events, when many forms of life disappeared and were replaced by forms that evolved from the survivors.

As we’ve seen, the fossil record provides a sweeping overview of the history of life over geologic time. Here we will focus on a few major events in that history, returning to study the details in Unit Five.

Figure 25.8 will help you visualize how long ago

these key events occurred against the vast backdrop of geologic time.

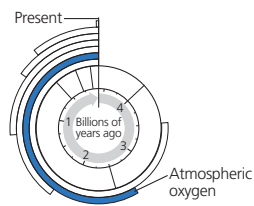
The First Single-Celled Organisms



Earth’s first organisms were single-celled prokaryotes that lived in the ocean. The earliest direct evidence of these organisms, dating from 3.5 billion years ago, comes from fossilized stromatolites.

Stromatolites are layered rocks that form when certain prokaryotes bind thin films of sediment together. Stromatolites and other early prokaryotes were Earth’s sole inhabitants for about 1.5 billion years. As we will see, these prokaryotes transformed life on our planet.

Photosynthesis and the Oxygen Revolution

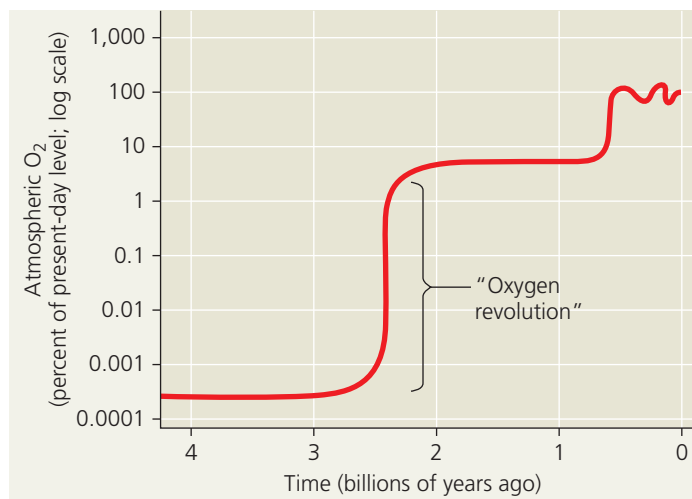


Most atmospheric oxygen gas (O_2) is of biological origin, produced during the water-splitting step of photosynthesis. When oxygenic photosynthesis first evolved—in photosynthetic prokaryotes similar to today's cyanobacteria—the free O_2

it produced probably dissolved in the surrounding water until it reached a high enough concentration to react with elements dissolved in water, including iron. This would have caused the iron to precipitate as iron oxide, which accumulated as sediments. These sediments were compressed into banded iron formations, red layers of rock containing iron oxide that are a source of iron ore today. Once all of the dissolved iron had precipitated, additional O_2 dissolved in the water until the seas and lakes became saturated with O_2 . After this occurred, the O_2 finally began to “gas out” of the water and enter the atmosphere. This change left its mark in the rusting of iron-rich terrestrial rocks, a process that began about 2.7 billion years ago.

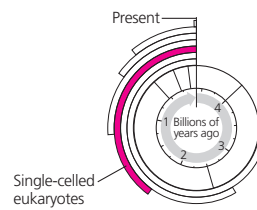
As shown in **Figure 25.9**, the amount of atmospheric O_2 increased gradually from about 2.7 to 2.4 billion years ago, but then shot up relatively rapidly to between 1% and 10% of its present level. This “oxygen revolution” had an enormous impact on life. In some of its chemical forms, oxygen attacks chemical bonds and can inhibit enzymes and damage cells. As a result, the rising concentration of atmospheric O_2 probably doomed many prokaryotic groups. Some species survived in habitats that remained anaerobic, where we find their descendants living today (see Concept 27.4). Among other survivors, diverse adaptations to the changing atmosphere evolved, including cellular respiration, which uses O_2 in the process of harvesting the energy stored in organic molecules.

▼ **Figure 25.9** The rise of atmospheric oxygen. Chemical analyses of ancient rocks have enabled this reconstruction of atmospheric oxygen levels during Earth's history.



The rise in atmospheric O_2 levels left a huge imprint on the history of life. A few hundred million years later, another fundamental change occurred: the origin of the eukaryotic cell.

The First Eukaryotes



The oldest widely accepted fossils of eukaryotes are of single-celled organisms that lived 1.8 billion years ago. Recall that eukaryotic cells have more complex organization than prokaryotic cells:

Eukaryotic cells have a nuclear envelope, mitochondria, endoplasmic reticulum, and other internal structures that prokaryotes lack. Also, unlike prokaryotic cells, eukaryotic cells have a well-developed cytoskeleton, a feature that enables eukaryotic cells to change their shape and thereby surround and engulf other cells.

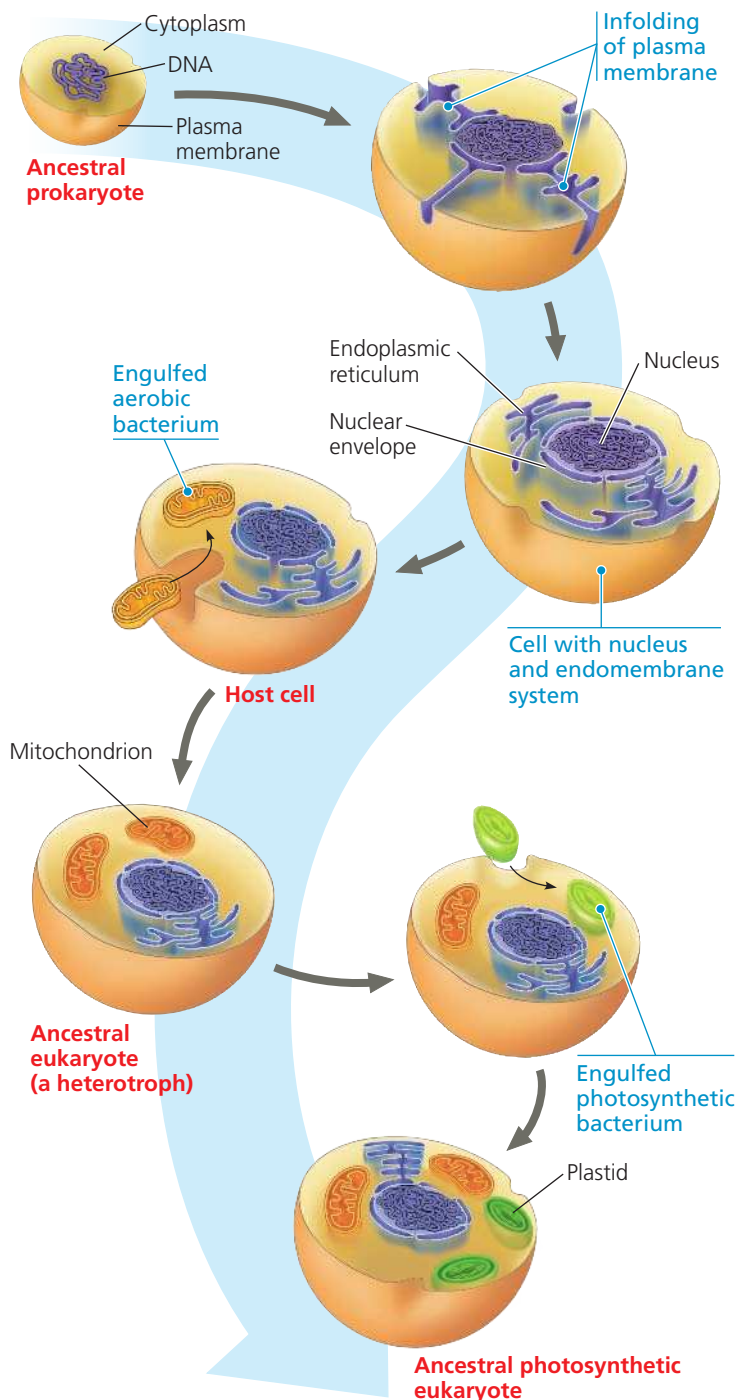
How did the eukaryotes evolve from their prokaryotic ancestors? Current evidence indicates that the eukaryotes originated by **endosymbiosis** when a prokaryotic cell engulfed a small cell that would evolve into an organelle found in all eukaryotes, the mitochondrion. The small, engulfed cell is an example of an *endosymbiont*, a cell that lives within another cell, called the *host cell*. The prokaryotic ancestor of the mitochondrion probably entered the host cell as undigested prey or an internal parasite. Though such a process may seem unlikely, scientists have directly observed cases in which endosymbionts that began as prey or parasites developed a mutually beneficial relationship with a host in as little as five years.

By whatever means the relationship began, we can hypothesize how the symbiosis could have become beneficial. For example, in a world that was becoming increasingly aerobic, a host that was itself an anaerobe would have benefited from endosymbionts that could make use of the oxygen. Over time, the host and endosymbionts would have become a single organism, its parts inseparable. Although all eukaryotes have mitochondria or remnants of these organelles, they do not all have plastids (a general term for chloroplasts and related organelles). Thus, the **serial endosymbiosis** hypothesis supposes that mitochondria evolved before plastids through a sequence of endosymbiotic events. As shown in **Figure 25.10**, both mitochondria and plastids are thought to have descended from bacterial cells. The original host—the cell that engulfed the bacterium whose descendants gave rise to the mitochondrion—is thought to have been an archaean or a close relative of the archaea.

A great deal of evidence supports the endosymbiotic origin of mitochondria and plastids:

- The inner membranes of both organelles have enzymes and transport systems that are homologous to those found in the plasma membranes of living bacteria.

▼ **Figure 25.10 A hypothesis for the origin of mitochondria and plastids through serial endosymbiosis.** The proposed host cell was an archaean or a close relative of the archaea. The proposed ancestors of mitochondria were aerobic, heterotrophic bacteria, while the proposed ancestors of plastids were photosynthetic bacteria. In this figure, the arrows represent change over evolutionary time.



➔ Mastering Biology Figure Walkthrough

- Mitochondria and plastids replicate by a splitting process that is similar to that of certain bacteria. In addition, each of these organelles contains circular DNA molecules

that, like the chromosomes of bacteria, are not associated with histones or large amounts of other proteins.

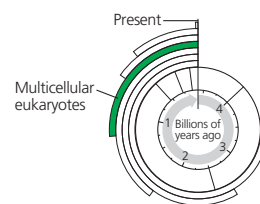
- As might be expected of organelles descended from free-living organisms, mitochondria and plastids also have the cellular machinery (including ribosomes) needed to transcribe and translate their DNA into proteins.
- Finally, in terms of size, RNA sequences, and sensitivity to certain antibiotics, the ribosomes of mitochondria and plastids are more similar to bacterial ribosomes than they are to the cytoplasmic ribosomes of eukaryotic cells.

In Concept 28.1, we'll return to the origin of eukaryotes, focusing on what genomic data have revealed about the prokaryotic lineages that gave rise to the host and endosymbiont cells.

The Origin of Multicellularity

An orchestra can play a greater variety of musical compositions than a violin soloist can; the increased complexity of the orchestra makes more variations possible. Likewise, the origin of structurally complex eukaryotic cells sparked the evolution of greater morphological diversity than was possible for the simpler prokaryotic cells. After the first eukaryotes appeared, a great range of unicellular forms evolved, giving rise to the diversity of single-celled eukaryotes that continue to flourish today. Another wave of diversification also occurred: Some single-celled eukaryotes gave rise to multicellular forms, whose descendants include a variety of algae, plants, fungi, and animals.

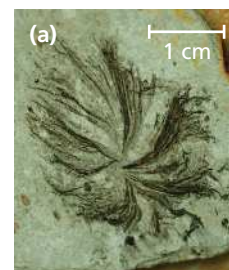
Early Multicellular Eukaryotes



Fossil evidence and DNA sequence data suggest that multicellular eukaryotes emerged about 1.3 billion years ago. The oldest known fossils of multicellular eukaryotes that can be resolved taxonomically are of relatively small red algae that lived 1.2 billion years ago. Larger and more diverse multicellular eukaryotes do not appear in the fossil record until about 600 million years ago (**Figure 25.11**). These fossils, referred to as the

▼ Figure 25.11 Life becomes large.

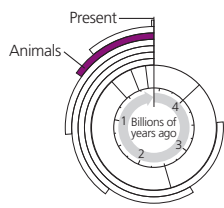
Fossils from the Ediacaran period document the oldest known macroscopic eukaryotes, such as (a) *Doushantuophyton*, an alga that lived 600 mya, and (b) *Kimberella*, a mollusc (or close relative) that lived 560 mya.



Ediacaran biota, were of mostly soft-bodied organisms—some over 1 m long—that lived from 635 to 541 million years ago (mya). The Ediacaran biota included both algae and animals, along with various organisms of unknown taxonomic affinity.

The rise of large eukaryotes in the Ediacaran period represents an enormous change in the history of life. Before that time, Earth was a microbial world: Its only inhabitants were single-celled prokaryotes and eukaryotes, along with an assortment of microscopic, multicellular eukaryotes. As the diversification of the Ediacaran biota came to a close about 541 million years ago, the stage was set for another, even more spectacular burst of evolutionary change: the “Cambrian explosion.”

The Cambrian Explosion

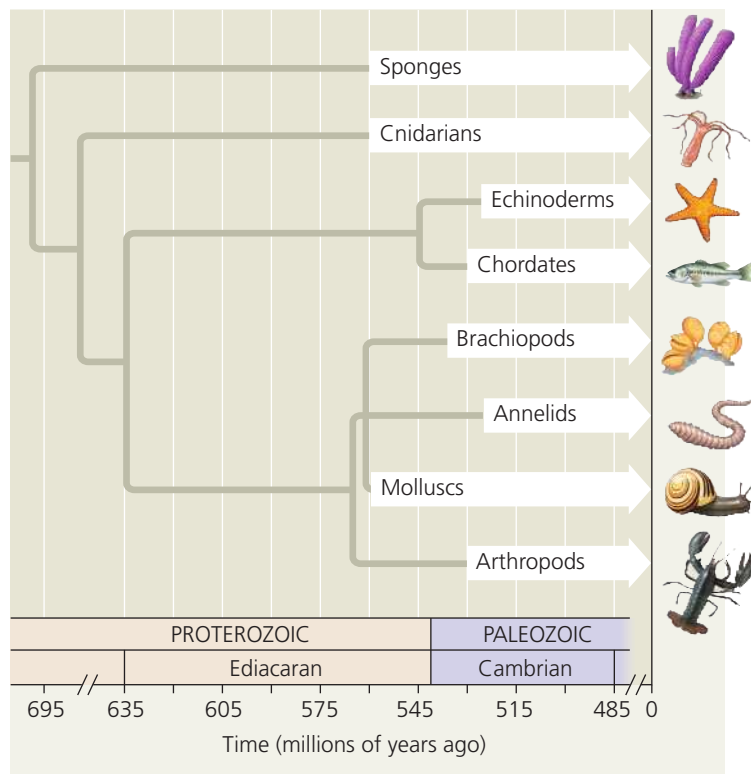


Many present-day animal phyla appear suddenly in fossils formed 535–525 million years ago, early in the Cambrian period. This phenomenon is referred to as the **Cambrian explosion**. Fossils of several animal groups—sponges, cnidarians (sea

anemones and their relatives), and molluscs (snails, clams, and their relatives)—appear in even older rocks dating from the late Proterozoic (**Figure 25.12**).

Prior to the Cambrian explosion, all large animals were soft-bodied. The fossils of large pre-Cambrian animals reveal little

▼ **Figure 25.12 Appearance of selected animal groups.** The white bars indicate the earliest fossil record of these animal groups.

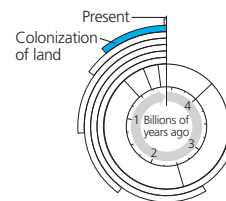


VISUAL SKILLS Circle the branch point that represents the most recent common ancestor of chordates and annelids. What is a minimum estimate of that ancestor's age?

evidence of predation. Instead, these animals appear to have been grazers (feeding on algae), filter feeders, or scavengers, not hunters. The Cambrian explosion changed all of that. In a relatively short period of time (10 million years), predators over 1 m in length emerged that had claws and other features for capturing prey; simultaneously, new defensive adaptations, such as sharp spines and heavy body armor, appeared in their prey.

Although the Cambrian explosion had an enormous impact on life on Earth, it appears that many animal phyla originated long before that time. Recent DNA analyses suggest that sponges had evolved by 700 million years ago; such analyses also indicate that the common ancestor of arthropods, chordates, and other animal phyla that radiated during the Cambrian explosion lived 670 million years ago. Researchers have unearthed 710-million-year-old sediments containing steroids indicative of a particular group of sponges—a finding that supports the molecular data. In contrast, the oldest macroscopic fossils of animals date from 560 million years ago and include *Kimberella*, shown in Figure 25.11. Overall, fossils and DNA analyses suggest that animals originated about 700 million years ago and then remained small for over 100 million years—until they diversified explosively during the Cambrian and beyond.

The Colonization of Land



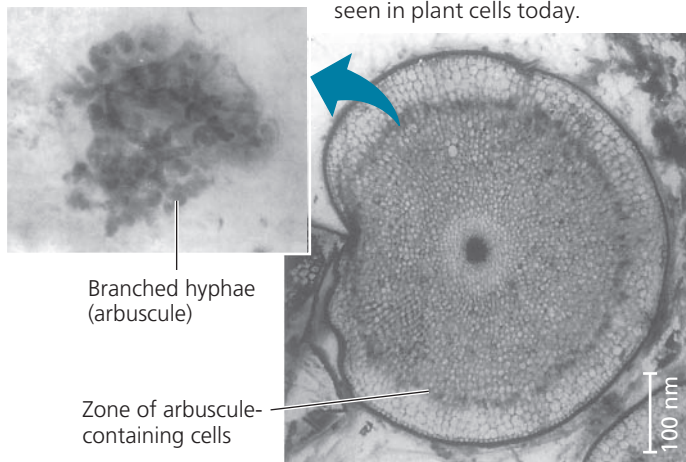
The colonization of land was another milestone in the history of life. There is fossil evidence that some prokaryotes lived on terrestrial surfaces as early as 3.2 billion years ago. However, larger forms of life, such as fungi, plants, and animals, did not begin to colonize land

until about 500 million years ago. This gradual evolutionary venture out of aquatic environments was associated with adaptations that made it possible to reproduce on land and that helped prevent dehydration. For example, many plants today have a vascular system for transporting materials internally and a waterproof coating of wax on their leaves that slows the loss of water to the air. Early signs of these adaptations were present 420 million years ago, at which time small plants (about 10 cm high) existed that had a vascular system but lacked true roots or leaves. By 40 million years later, plants had diversified greatly and included trees and many other plants with true roots and leaves.

Plants appear to have colonized land in the company of fungi. Even today, the roots of most plants are associated with fungi that aid in the absorption of water and minerals from the soil (see Concept 31.1). These root fungi (or *mycorrhizae*), in turn, obtain their organic nutrients from the plants. Such mutually beneficial associations of plants and fungi are evident in some of the oldest fossilized plants, dating this relationship back to the early spread of life onto land (**Figure 25.13**).

Although many animal groups are now represented in terrestrial environments, the most widespread and diverse land

▼ **Figure 25.13 An ancient symbiosis.** This 405-million-year-old fossil stem (cross section) documents mycorrhizae in the early plant *Aglaophyton major*. The inset shows an enlarged view of a cell containing a branched fungal structure called an arbuscule; the fossil arbuscule resembles those seen in plant cells today.



animals are arthropods (particularly insects and spiders) and tetrapods. Arthropods were among the first animals to colonize land, roughly 450 million years ago. The earliest tetrapods found in the fossil record lived about 365 million years ago and appear to have evolved from a group of lobe-finned fishes (see Concept 34.3). Tetrapods include humans, although we are late arrivals on the scene. The human lineage diverged from other primates around 6–7 million years ago, and our species originated only about 195,000 years ago. If the clock of Earth’s history were rescaled to represent an hour, humans appeared less than 0.2 second ago.

➔ **Mastering Biology**
Interview with Geerat Vermeij: What fossils reveal about the history of life



CONCEPT CHECK 25.3

1. The first appearance of free oxygen in the atmosphere likely triggered a massive wave of extinctions among the prokaryotes of the time. Why?
2. What evidence supports the hypothesis that mitochondria preceded plastids in the evolution of eukaryotic cells?
3. **WHAT IF?** What would a fossil record of life today look like?

For suggested answers, see Appendix A.

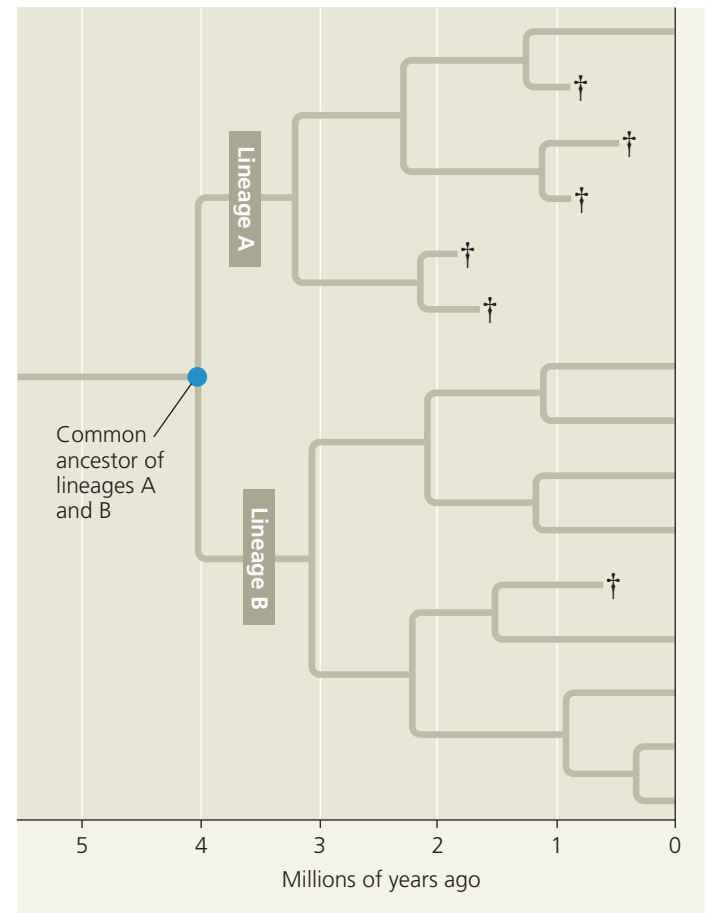
CONCEPT 25.4

The rise and fall of groups of organisms reflect differences in speciation and extinction rates

From its beginnings, life on Earth has been marked by the rise and fall of groups of organisms. Anaerobic prokaryotes originated, flourished, and then declined as the oxygen content of the atmosphere rose. Billions of years later, the first tetrapods emerged from the sea, giving rise to several major new groups of organisms. One of these, the amphibians, went

▼ **Figure 25.14 How speciation and extinction affect diversity.**

The species diversity of an evolutionary lineage will increase when more new member species originate than are lost to extinction. In this hypothetical phylogenetic tree, each horizontal branch represents one species. By 2 million years ago both lineage A and lineage B have given rise to four species, and no species have become extinct (denoted by a dagger symbol, †). By time 0, however, lineage A contains only one species, while lineage B contains eight species.



VISUAL SKILLS Consider the period from 2 million years ago to time 0. For each lineage, determine how many speciation and extinction events occurred during that time.

on to dominate life on land for 100 million years, until other tetrapods (including dinosaurs and, later, mammals) replaced them as the dominant terrestrial vertebrates.

The rise and fall of these and other major groups of organisms have shaped the history of life. Narrowing our focus, we can also see that the rise or fall of any particular group is related to the speciation and extinction rates of its member species (**Figure 25.14**). Just as a population increases in size when there are more births than deaths, the rise of a group of organisms occurs when more new species are produced than are lost to extinction. The reverse occurs when a group is in decline. In the **Scientific Skills Exercise**, you can interpret data from the fossil record about changes in a group of snail species in the early Paleogene period. Such changes in the fates of groups of organisms have been influenced by large-scale processes such as plate tectonics, mass extinctions, and adaptive radiations.